# Network Estimation Workbook

**The theory behind GnssProcessor's network processor, in 41 questions**

October 2026

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## How the course works

The course teaches the estimation theory of the network processor and its reference choices (datum, S-basis) from this project's own record. It starts on familiar ground, one receiver, and adds one idea at a time. Lessons 1 to 3 work the theory by hand on small examples, then find the same structure in a real run. Lessons 4 to 7 apply it to the project's own choices: the four reference choices, what a user receives, the satellite clock and the integer ambiguities. Lessons 8 and 9 show how the results are tested on simulated and real data and what they support. Every question names the earlier questions it builds on, so the lessons are best worked in order.

Three kinds of question appear in every lesson: DERIVE (a calculation by hand), READ (what a given line of a card, configuration file, log or source file says) and MEASURE (run an existing tool on

an existing run and interpret the number). The last question of the course, SYNTHESIS, gathers what the presentation of 29 October may claim.

1. Work one lesson per sitting, in order.
2. Read the lesson's opening section and figures; open a source only when a question sends there.
3. Write each answer before turning to the answer key. An answer is complete when it names its source.
4. Turn to the key in two stages. Read only the answer's *Strategy* first and try again; then read the *Solution*, which explains each step, *Why it works*, the usual *Pitfall* and the *Takeaway* that later lessons build on (each question links to its answer and back).
5. Where an answer differs, rerun the key's check.

| | Lesson | What it builds | Questions |
|---|---|---|---|
| 1 | One receiver | the three directions a receiver's data cannot fix | 3 |
| 2 | Two receivers | rank, a reference choice, what it changes and what it cannot, a user, a change of satellites | 6 |
| 3 | The real network | elimination of the stations, two constellations, the count of Eq. 24, the parameter table, the rank gate | 5 |
| 4 | The project's reference choices | the theorem, the four choices and their rows, the code-bias datum, the station level | 5 |
| 5 | What a user sees | what a user absorbs, the header line of the corrections, a declared step, the reference satellite of a double difference | 4 |
| 6 | The satellite clock | the anchor as a step and as a time constant, the hold, two postures, the cost of leaving the final clock | 5 |
| 7 | Ambiguities | why an integer is mostly ionosphere, cycles, integer and float pins, the oracle pin, the corrections at the null | 5 |
| 8 | Simulated data | what a simulation tests, where a run starts, grading in the network's basis, the user's floor | 4 |
| 9 | Real data | the test station and 16 stations, five days, the clock against CODE and GFZ, the claims | 4 |

**Before the measure questions.** Commands run from `C:/GnssProcessor` in Git Bash, with `C:/Python313/python.exe` (the Python on the path lacks numpy). Several runs and the clock data set are on the D: drive (`D:/GnssProcessorSupport/runs_nom/`, `D:/GnssProcessorSupport/fixtures/`), which must be connected. No question builds the code or starts a new run; all commands together take about two minutes. The status report is `D:/GnssProcessorSupport/presentation/status_update_2026-10-29/status_update_2026-10-29.tex`; paths starting `results/` or `figures/deck/` in captions are in that folder.

**Terms.** The course uses plain words; the cards, code and configuration files use the record's labels.

| In this workbook | In the record | Meaning |
|---|---|---|
| our network, the network processor | producer | `runNetwork`: estimates satellite clocks and biases from many stations |
| corrections, the correction file | the wire, `products.txt` | what the network writes per epoch: C (clock), B (code bias), P (phase bias) and other records |
| user | consumer | a receiver positioned with the corrections |
| test station | held-out station | a reference station left out of the network and positioned as a user: WTZZ00DEU (real), SZZ000SIM (simulated) |
| 16 further stations | census | users positioned afterwards from the stored corrections |
| real data | field | 58 IGS stations, 8 May 2024 (day 129) |
| simulated data set | fixture, sim bed | `runs/simnet60_quiet_col_24h` (quiet), `simnet60_clk_col_24h` (clock error) |
| reference configuration | golden | `config/golden/golden_field_refound.json` and `golden_sim_refound.json`, generated |
| case, setting | arm, knob | a copy of a reference configuration with one setting changed; a configuration key |
| the first hour | ignition | hour 1 after a cold start |
| reference choice | datum, S-basis, gauge | the constraints that fix the directions no data determine |
| header line of reference choices | S line | line 2 of `products.txt` |
| perfect corrections | truth wire | corrections written from the simulator's truth |

# 1 One receiver: three directions the data cannot fix

**Goal.** Start from the familiar user model and find the first null directions: combinations of unknowns along which every observation stays the same.

**The idea.** For receiver $r$, satellite $i$ and signal $s$ on two frequencies, the code and phase equations of the status report (section "One observation model") are

$$P^i_{r,s} = \rho^i_r + c(\mathrm{d}t_r - \mathrm{d}t^i) + T^i_r + \mu_s I^i_r + B_{r,s} - B^i_s + \varepsilon, \tag{1}$$

$$\Phi^i_{r,s} = \rho^i_r + c(\mathrm{d}t_r - \mathrm{d}t^i) + T^i_r - \mu_s I^i_r + \lambda_s N^i_{r,s} + b_{r,s} - b^i_s + \varepsilon, \tag{2}$$

with $\mu_s = f_1^2/f_s^2$ (so $\mu_1 = 1$), $I^i_r$ the slant ionospheric delay on the first frequency, $B$ and $b$ the code and phase biases, and $N$ the ambiguity in cycles. A PPP user takes the orbit and the satellite terms $\mathrm{d}t^i$, $B^i_s$ and $b^i_s$ from a product and estimates the rest. Even then some combinations of the remaining unknowns leave every observation unchanged. Such a combination is a null direction of the design matrix: the data cannot fix a parameter along it, so a convention must. The rest of the course follows these directions from one receiver into a network.

**Question 1.1** DERIVE **The receiver clock and the code biases**

answer, p. 23

1. Show that the change $c\,\mathrm{d}t_r \to c\,\mathrm{d}t_r + \delta$, with every receiver bias $B_{r,1}, B_{r,2}, b_{r,1}, b_{r,2}$ changed by $-\delta$, leaves all four rows of a satellite unchanged.
2. Write the two receiver code biases as $B_{r,s} = B_{r,\mathrm{IF}} + \mu_s\, d_r$. Solve for $B_{r,\mathrm{IF}}$ and $d_r$ in terms of $B_{r,1}$ and $B_{r,2}$, and show that $B_{r,\mathrm{IF}}$ is their ionosphere-free combination.

3. Rewrite the code rows with this split. Which combination of clock and code biases do they contain, and what is a user's estimated "receiver clock" therefore?

**Question 1.2** DERIVE **The phase bias and the ambiguity**

*Builds on 1.1.* answer, p. 24

1. Show that, for each signal $s$, the change $b_{r,s} \to b_{r,s} - \delta_s$ together with $N^i_{r,s} \to N^i_{r,s} + \delta_s/\lambda_s$ for every satellite leaves every phase row unchanged. How many such directions does one receiver have?
2. A PPP user estimates one float ambiguity per satellite and signal and no receiver phase bias. What does that float ambiguity contain, and why is it not an integer even with exact satellite phase biases?
3. What does a difference between two satellites at the same receiver remove, and what must the product supply for that difference to be an integer?

**Question 1.3** DERIVE **The ionosphere and the geometry-free code bias**

*Builds on 1.1 and 1.2.* answer, p. 24

1. With the split of Question 1.1, show that the code rows of every satellite contain the slant ionosphere only as $I^i_r + d_r$.
2. Show that $I^i_r \to I^i_r + \delta$ for every satellite, with $d_r \to d_r - \delta$, leaves the code rows unchanged, and find what absorbs the change on the phase rows.
3. What does a user's estimated slant ionosphere therefore contain? What would have to be fixed by convention for it to be the physical ionosphere?

**What this lesson establishes.**

- One receiver has three kinds of null direction: the clock against the ionosphere-free code bias; each signal's phase bias against all of that signal's ambiguities; all slant ionospheres against the geometry-free code bias.
- A parameter estimated along such a direction is a combination whose split a convention fixes: here the clock takes $B_{r,\mathrm{IF}}$, the ambiguities take the phase bias, the ionosphere takes $d_r$.
- A network has the same directions at every receiver, and their mirror images at every satellite (Section 2). For the meeting: these directions are why a network must state its reference choices.

## 2 Two receivers: rank, reference choice and invariance

**Goal.** Work the smallest network by hand: find its null directions, remove them with a reference choice, and see what the choice changes and what it cannot change.

**The idea.** The Mathematical Reference works this example (`docs/report/network_example.tex`, "Worked example: estimability in a two-receiver network"). Receivers $A$ and $B$ observe satellites 1 to 4 (Figure 1). Positions are known; one frequency; no troposphere or ionosphere; no noise. The code collapses to a difference of two clock lumps,

$$P^i_r = a_r - g^i, \qquad a_r = c\,\mathrm{d}t_r + B_r, \qquad g^i = c\,\mathrm{d}t^i + B^i, \tag{3}$$

which is the convention of Question 1.1 already applied: each clock carries its code bias. The phase is

$$\Phi_r^i = a_r - g^i + b^i + N_r^i, \tag{4}$$

with $b^i$ the satellite phase bias and $N_r^i$ the ambiguity in metres, which also absorbs the receiver's phase bias (Question 1.2). The truth, in metres:

$$a_A = 5, \quad a_B = 3; \qquad g = (2, -1, 4, 0.5); \qquad b = (0.10, -0.20, 0.05, 0.15);$$
$$N_A = (10, -5, 20, 8), \qquad N_B = (12, -3, 18, 9).$$

| $P_r^i$ | 1 | 2 | 3 | 4 | $\Phi_r^i$ | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|---|---|---|---|
| $A$ | 3 | 6 | 1 | 4.5 | $A$ | 13.10 | 0.80 | 21.05 | 12.65 |
| $B$ | 1 | 4 | −1 | 2.5 | $B$ | 13.10 | 0.80 | 17.05 | 11.65 |

A third receiver, $C$, appears in Question 2.5 as a user of the network's corrections.

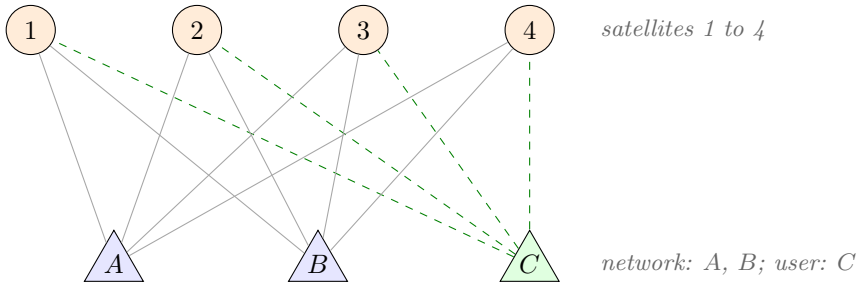



Figure 1: The worked example: two network receivers and four satellites; the user $C$ of Question 2.5 applies the network's corrections.

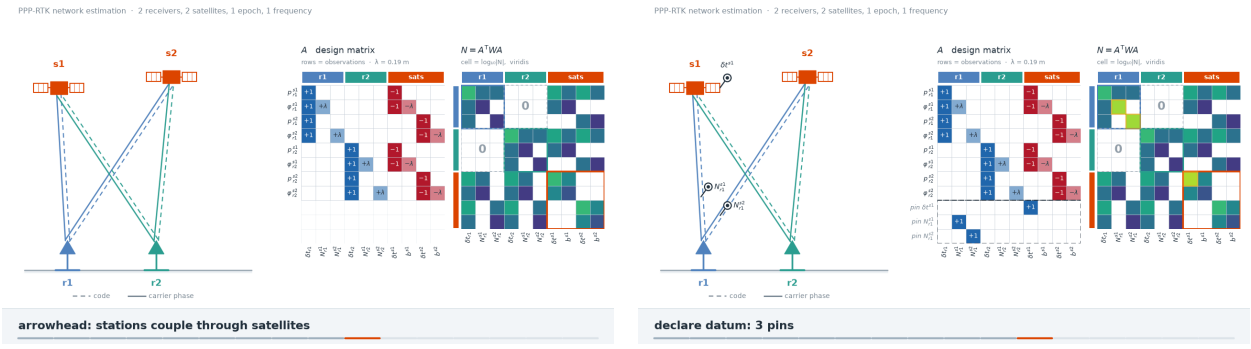



Figure 2: The same network with two satellites, from the project's concept animation (`D:/GnssProcessorSupport/presentation/figures/network_concept.gif`): design matrix $\boldsymbol{A}$ and normal matrix $\boldsymbol{N} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{W}\boldsymbol{A}$. Left: the empty blocks show that the receivers meet only through satellite terms. Right: eight observations, ten unknowns, three null directions, removed by three pins (the clock of satellite 1 and receiver $A$'s two ambiguities).

### Question 2.1 DERIVE The normal matrix and its null vector

*Builds on 1.1.* answer, p. 25

Order the unknowns $\boldsymbol{x} = [a_A, a_B, g^1, g^2, g^3, g^4]^{\mathrm{T}}$.

1. Write the $8 \times 6$ design matrix of the code rows and the normal matrix $\boldsymbol{N} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ (unit weights).
2. Show that $\boldsymbol{N}\mathbf{1} = \mathbf{0}$ for the all-ones vector. What is the rank of $\boldsymbol{N}$?
3. What change of the unknowns does $\mathbf{1}$ describe, and why can no number of observations of this kind fix it?

**Question 2.2** DERIVE **Two reference choices, one fit**

*Builds on 2.1.* answer, p. 25

1. Solve the code rows with the pin $a_A \equiv 0$, then with the pin $g^1 \equiv 0$.
2. Subtract the two solutions. What is the difference, in the terms of Question 2.1?
3. Compute $a_A - a_B$, $g^2 - g^1$, $g^3 - g^1$ and $a_B - g^4$ in both solutions. Why do they agree?
4. Under $a_A \equiv 0$, what does the estimated clock of satellite 2 equal in terms of the true clocks?

**Question 2.3** DERIVE **Phase: the satellite phase bias needs a second receiver**

*Builds on 2.2.* answer, p. 26

1. With code and phase together, count the unknowns, the observations, the null directions and the redundancy. Which null directions does the phase add?
2. With receiver $A$ alone, what does each satellite's phase determine, and which pair of unknowns can it not separate? What changes with receiver $B$?
3. Apply the reference choice $a_A \equiv 0$ with $N_A^i \equiv 0$ for every satellite. Compute the emitted phase bias $\bar{b}^i$ and the estimate $\hat{N}_B^i$ for the four satellites.
4. The physical phase bias of satellite 1 is 0.10 m and the emitted one 10.10 m. What did it absorb? What is $\hat{N}_B^i$?

**Question 2.4** DERIVE **Where the hardware biases went**

*Builds on 1.1, 1.2 and 2.3.* answer, p. 27

Write the network with every hardware bias explicit: receiver and satellite code biases $B_r$, $B^i$, receiver and satellite phase biases $b_r$, $b^i$, and the integer ambiguities. For $f$ frequencies, $n$ receivers and $m$ satellites, Odijk et al. [2] count six types of null direction (`SBASIS-THEORY-E.md`, section E2): 1a, receiver clock against satellite clock, 1; 1b, receiver against satellite hardware biases, $2f$; 2a, receiver clock against receiver biases, $n-1$; 3a, satellite clock against satellite biases, $m$; 4, receiver phase bias against ambiguities, $f(n-1)$; 5, satellite phase bias against ambiguities, $f\,m$.

1. Evaluate the six types for $f = 1$, $n = 2$, $m = 4$ and add them.
2. The worked example defined $a_r$, $g^i$ and $N_r^i$ as lumps. How many unknowns did the lumps remove, and how does the rank change? Which types are gone and which remain? Check the remainder against Question 2.3.

**Question 2.5** DERIVE **A user of the emitted corrections**

*Builds on 2.2 and 2.3.* answer, p. 27

A third receiver $C$, not in the network, has the truth $a_C = 4$ and $N_C = (11, -4, 19, 10)$ m.

1. Compute its four code and four phase observations.

2. $C$ applies the network's satellite clocks $\hat{g}^i$ (from the choice $a_A \equiv 0$) and phase biases $\bar{b}^i$. Solve for its clock $\hat{a}_C$ and its ambiguities $\tilde{N}^i_C$.
3. What do $\hat{a}_C$ and $\tilde{N}^i_C$ equal in terms of the true values? Whose values does $C$'s solution carry, and does $C$ need to know it?

**Question 2.6** DERIVE **When the satellite set changes**

*Builds on 2.2.* answer, p. 27

1. From the solution with $g^1 \equiv 0$, give the solution under the sum-to-zero choice $\sum_i g^i = 0$ over the four satellites.
2. Satellite 4 sets. Recompute both solutions with satellites 1 to 3. By how much does every clock move under each choice?
3. Instead satellite 1, the pinned one, sets, and the pin moves to satellite 2. By how much does every clock move?
4. Does any of these changes move an estimable quantity? What must a user be told at such an epoch, and why?

**What this lesson establishes.**

- A reference choice is a set of pins or rows that removes the null directions; every admissible choice fits the data equally well.
- Differences (estimable functions) agree under every choice; the emitted values carry the choice (the satellite clock carries the reference receiver's clock, the phase bias its ambiguity).
- A user applying the corrections inherits the network's reference without knowing it: its estimates are relative to the network.
- A sum-to-zero reference moves whenever a satellite rises or sets; a pin moves only when its own satellite leaves. Either move must be declared.

# 3 The real network: elimination, counting and the rank gate

**Goal.** Scale the worked example up to the real network: eliminate the stations, count the null directions, and find them again in a run.

**The idea.** The network processor solves one joint system per epoch over all stations (status report, step 6). The unknowns shared by all stations, $\boldsymbol{x}_\mathrm{B}$, are the satellite clocks, code biases and phase biases; each station $r$ has its own, $\boldsymbol{x}_r$: ambiguities, slant ionosphere, zenith wet delay, receiver clock, inter-system bias and receiver code-bias difference. Stations meet only through $\boldsymbol{x}_\mathrm{B}$, so the normal matrix is an arrowhead (Figure 2, left) and each station's block is eliminated exactly before the shared solve:

$$\bar{\boldsymbol{N}}_\mathrm{BB} = \sum_r \left( \boldsymbol{N}^{(r)}_\mathrm{BB} - \boldsymbol{N}_{\mathrm{B}r} \boldsymbol{N}_{rr}^{-1} \boldsymbol{N}_{\mathrm{B}r}^\mathrm{T} \right) + \boldsymbol{N}^\mathrm{prior}_\mathrm{BB}, \qquad \boldsymbol{x}_r = \boldsymbol{N}_{rr}^{-1} \big( \boldsymbol{u}_r - \boldsymbol{N}_{\mathrm{B}r}^\mathrm{T} \boldsymbol{x}_\mathrm{B} \big). \tag{5}$$

From epoch to epoch the solution is a Kalman filter in information form. Every epoch a check, the *rank gate*, counts the eigenvalues of $\bar{\boldsymbol{N}}_\mathrm{BB}$ below `rank_gate_tol` $\times \lambda_\mathrm{max}$ and requires exactly the declared number (`include/network/solve/Sbasis.hpp`, `src/core/network/solve/RankGate.cpp`). A single-receiver PPP filter has the same shape, with epochs in the place of stations (Figure 3).

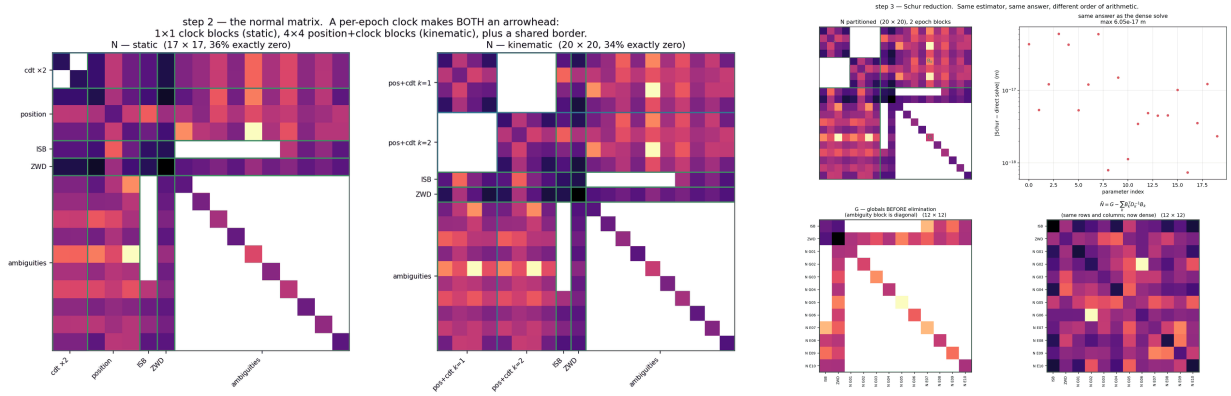



Figure 3: Left: one receiver over two epochs; a clock per epoch makes $1 \times 1$ (static) or $4 \times 4$ (moving receiver) epoch blocks around a shared border. Right: eliminating the blocks first gives the direct solution to $6 \times 10^{-17}$ m. September 2026 illustrations.

### Question 3.1 DERIVE **Eliminating the receivers**

*Builds on 2.1 and 2.2.* answer, p. 28

In the code system of Question 2.1, the receiver clocks are the local unknowns and the satellite clocks the shared ones.

1. Compute the reduced shared normal $\bar{\boldsymbol{N}}_{gg} = \boldsymbol{N}_{gg} - \boldsymbol{N}_{ga}\boldsymbol{N}_{aa}^{-1}\boldsymbol{N}_{ag}$ and the reduced right-hand side $\bar{\boldsymbol{u}}_g = \boldsymbol{u}_g - \boldsymbol{N}_{ga}\boldsymbol{N}_{aa}^{-1}\boldsymbol{u}_a$ from the pseudorange table.
2. Find the eigenvalues of $\bar{\boldsymbol{N}}_{gg}$ and its normalized null eigenvector. What weight does it put on each satellite clock?
3. Solve the reduced system with $g^1 \equiv 0$, then back-substitute the receiver clocks with Equation (5). Compare with Question 2.2.

### Question 3.2 DERIVE **Two constellations**

*Builds on 3.1.* answer, p. 29

Add Galileo satellites. Each receiver gets an inter-system bias $d_r^E$ on its Galileo rows ($d_r^G \equiv 0$ by definition: GPS is the reference system), so $P_r^i = a_r + d_r^{g(i)} - g^i$.

1. Find the null directions of this system.
2. After the receiver clocks and inter-system biases are eliminated, how many null directions does the reduced shared normal keep, and what do they look like on the satellite clocks?
3. For 32 GPS and 24 Galileo satellites, what weight does each normalized direction put on each clock?

### Question 3.3 DERIVE **Eq. 24 for the real network**

*Builds on 1.3 and 2.4.* answer, p. 29

1. Sum the six types of Question 2.4 to obtain Eq. 24 of [2], $1 + 2f + (1 + f)(n - 1 + m)$. Add the slant-ionosphere term $n - 1 + m$ (the direction of Question 1.3, now at every receiver and every satellite).
2. Figure 4 uses one constellation of 32 satellites and two frequencies with a slant ionosphere. Check its 369 directions for 60 stations, 361 for 58 and 1,349 for 305.

3. The first epoch of the simulated reference run has $n = 60$, $m = 56$ (32 GPS, 24 Galileo) and $f = 2$. What does Eq. 24 with the slant term give?

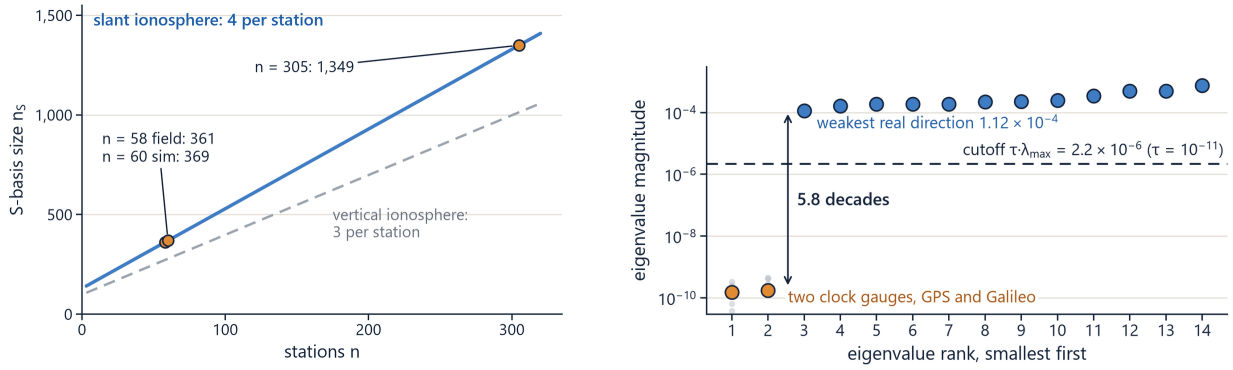



Figure 4: Left: Eq. 24 against the number of stations, one constellation of 32 satellites, two frequencies: about four more directions per station with a slant ionosphere, three with a vertical one (deck slide 6). Right: the rank gate on the first real-data epoch, the 14 smallest eigenvalues of the $414 \times 414$ reduced shared normal matrix: two near $10^{-10}$ (one clock direction per constellation), the weakest real direction 5.8 decades higher, the dashed line the cutoff (deck slide 7).

### Question 3.4 READ The parameters as armed

*Builds on 1.1 to 1.3 and 3.1.* answer, p. 30

Print the table of the simulated reference run (60 stations, GPS and Galileo, two signals each):

```
sed -n '658,686p' runs/_rf0hv15_sim6/stdout.txt
```

1. Map each unknown of Equations (1) and (2) to a row of the table. Which term has no row, and where did Question 1.2 put it?
2. How many shared and how many local columns does the run carry at its first epoch?

### Question 3.5 MEASURE The rank gate on the real network

*Builds on 3.2 and 3.3.* answer, p. 30

Run (about 9 s):

```
C:/Python313/python.exe -I scripts/rf_null_space_read.py runs/_rf0hv15_sim6
grep -n "rank gate evaluated" runs/_rf0hv15_sim6/stdout.txt
```

1. Which two eigenvalues fall below the cutoff at the first epoch, and what weights do their eigenvectors carry? Compare with Question 3.2.
2. How many decades lie between the cutoff and the larger of the two, and between the cutoff and the weakest real direction? Does that meet the placement rule of `src/core/network/solve/RankGate.cpp` (lines 408 to 410)?
3. Eq. 24 gave 465 directions; the gate declares 2. Where did the others go? (Question 2.4 holds the key; Question 4.3 counts the rows.)

**What this lesson establishes.**

- Eliminating the stations is exact; the reduced shared normal keeps the network's shared null directions.
- With two constellations the shared null space is two flat clock shifts, weight $1/\sqrt{m_g}$ on every clock of system $g$; the real run shows exactly these, at 0.1767 and 0.2040.
- Eq. 24 counts hundreds of directions; lumps, pins and priors remove almost all of them before the gate, which sees one per constellation. For the meeting: the per-epoch check confirms this count on every epoch.

# 4 The project's four reference choices

**Goal.** Name the reference choices the reference configurations make, count their rows, and find the one direction they leave to a weak prior.

**The idea.** A reference choice is a set of constraints $\boldsymbol{B}^{\mathrm{T}}\boldsymbol{x} = \boldsymbol{0}$, or pins, that picks one solution among those that fit the data equally well. The S-transformation $\boldsymbol{S} = \boldsymbol{I} - \boldsymbol{V}[(\boldsymbol{S}^{\perp})^{\mathrm{T}}\boldsymbol{V}]^{-1}(\boldsymbol{S}^{\perp})^{\mathrm{T}}$ projects onto the estimable space along the null space $\boldsymbol{V}$; its rows are the estimable combinations, and the reference directions carry zero variance (`SBASIS-THEORY-E.md`, section E1, after [2, 3]). The record names four kinds of choice (section E3), each with an analogue in Section 2: a *pin* ($g^1 \equiv 0$); a *prioritized anchor* ($N_A^i \equiv 0$, but elected per satellite and signal); a *sum-to-zero* row ($\sum g^i = 0$); an *ensemble*, an inner constraint over all receivers, measured dead and deleted on 2026-10-14. The reference configurations combine four choices:

| Choice | What it fixes | Settings (`golden_sim_refound.json`) | First epoch, simulated |
|---|---|---|---|
| Clock pin | one satellite clock per constellation (the pivot); Questions 2.2 and 3.2 | no key (the default); `sticky_pivot`, `handover_stransform` | 2 pins (E02, G01) |
| Prioritized phase anchors, held basis | one ambiguity per satellite and signal, frozen at its basis value; Question 2.3 | `phase_datum_prioritized`, `phase_datum_held_basis`, `ambiguity_vehicle float_kf` | 112 pins on 18 stations |
| Code-bias datum | per satellite the ionosphere-free pin and the geometry-free pin at the broadcast $T_{GD}$; Questions 1.1 and 1.3 at the satellites | `estimate_sat_code_bias`, `sat_gf_datum broadcast_tgd`, `sat_gf_datum_time_linked` | $56 + 56$ rows |
| Zero-mean receiver DCB prior | the mean of the receivers' geometry-free code biases, $\sigma$ 10 m; Question 1.3 at the receivers | `rx_code_bias_datum`, `station_prior_before_pd` | 120 prior entries on 60 stations |

One direction per station and constellation is left to the weak prior: the station's *level*, its slant ionosphere, receiver geometry-free code biases and ambiguities shifted together. It is local, so it is eliminated before the rank gate sees anything.

### Question 4.1 READ The theorem in the record

*Builds on 2.2.* answer, p. 31

Read the boxed theorem of `SBASIS-THEORY-E.md`, section E1, and the paragraph "Datum-invariant even before choosing a datum" of section E2.

1. What exactly is invariant, and under which choices? Where did Question 2.2 show it in six unknowns?
2. Which of these are estimable before any choice: position, zenith delay, vertical ionosphere, a satellite clock, a satellite phase bias?
3. A differential satellite clock wander of about 47 cm was measured before the re-founding. Why could no change of reference cure it, and what kind of change can?

**Question 4.2** READ **Four choices in the reference configuration**

*Builds on 2.3, 2.6 and 4.1.* answer, p. 32

Confirm that the reference configurations are the generated ones, then read `config/golden/golden_sim_refound.json`:

```
C:/Python313/python.exe scripts/make_golden_config.py --verify
```

1. Through which keys does the file arm each choice of the table above?
2. How does each choice behave when a satellite rises or sets, and which of them, by Question 2.6, would move at every rise and set?
3. Which datum-looking keys are absent from the file?

**Question 4.3** DERIVE **The rows of the first epoch**

*Builds on 3.5 and 4.2.* answer, p. 32

The simulated case's first epoch has 56 satellites (32 GPS, 24 Galileo) at 60 stations, two signals per constellation; each satellite carries one clock, two phase biases and two code biases.

1. How many shared columns are there, and how many rows does each choice contribute?
2. Account for the gate's 2: which rows act before the gate, and which after?
3. What does the solver do on an epoch whose count differs, and what would S-basis completion add?

Check against lines 631, 679 to 738 and 4109 to 4112 of `D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/stdout.txt`.

**Question 4.4** MEASURE **The code-bias datum on the corrections**

*Builds on 1.1 and 1.3.* answer, p. 33

Read the first epoch's code-bias records of G05 in the simulated reference case and the G05 record of the broadcast navigation file:

```
grep -n -m2 '^B,G05,' D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/products.txt
sed -n '562,568p' data/2024-129/products/BRDC00IGS_R_20241290000_01D_MN.rnx
```

1. What is the ratio of the C2W record to the C1C record, what is their ionosphere-free combination, and how does the C1C record compare with $cT_{GD}$?
2. Where does the satellite's true geometry-free code bias go in the network solution, and what must a user do with its own slant ionosphere for the records to apply?

**Question 4.5** DERIVE **The direction the shared check cannot see**

*Builds on 1.3, 3.1 and 4.3.* answer, p. 34

Card R1D (section 1) gives the loadings of the GPS level direction of station MAJU00MHL: the receiver DCB carries a share 0.0074 of the unit direction, the band-5 code delay $0.2392^2$ and the second-channel offset $0.1334^2$. The receiver DCB prior has $\sigma = 10$ m.

1. What standard deviation does the prior alone give along the unit level direction? Check the card's "about 116 m".
2. A zero-mean row on the band-5 delay and the second-channel offset at the same $\sigma$ would add information. By what factor? Check "about eleven times".
3. Card ALD (section 3) gives $v^{\mathrm{T}}Nv = 110$ per unit level for the coupling rows. What $\sigma$ follows, and which rows hold the level?
4. Why can the rank gate never see this direction, and how did it carry the GPS L5 event into the corrections?

**What this lesson establishes.**

- No admissible reference choice changes an estimable value or its variance; only estimation (weights, dynamics, model) can.
- The project pins one clock per constellation, anchors one ambiguity per satellite and signal, pins each satellite's code biases to the broadcast group delay, and holds the receivers' code-bias mean with a weak prior.
- The published code biases of the pinned pair carry no estimate; the ionosphere absorbs the true geometry-free bias.
- Each station's level is held by the carried coupling, not by the datum, and the gate cannot see it. For the meeting: the choices are 226 rows and 120 prior entries at the first simulated epoch, not "a few".

# 5 What a user receives and sees

**Goal.** Follow the corrections to a user: what it receives, what it cannot see, and what may therefore be graded.

**The idea.** A user positions itself with the equations of Section 1, with the network's clock $C$, code bias $B$ and phase bias $P$ in place of the satellite terms, and with its own receiver clock, slant ionosphere and float ambiguities. The three null directions of Section 1, now at the user's receiver, absorb three parts of the corrections: a change common to all clocks of a system, a constant on a satellite's phase bias, and a geometry-free offset of a satellite's code biases. These are the directions a reference choice moves, so a user cannot see the choice (Question 2.5 showed it with numbers). The S-system theory says the same of the correction covariance: its rank-one clock and ionosphere parts "lie in the user's own design range and are absorbed" (`SBASIS-THEORY-E.md`, section E4). Hence standing invariant 5 of `CLAUDE.md`: grade only datum-invariant or single-difference quantities. The correction file declares its choices in a header line (the S line) and marks every change of pivot with an epoch record (`E`, `datum_changed=1`).

**Question 5.1** DERIVE **Three things a float user absorbs**

*Builds on 1.1 to 1.3 and 2.5.* answer, p. 35

Take the user's code and phase rows for satellite $i$ of system $g$, with receiver clock, inter-system bias, slant ionosphere $I^i$ and float ambiguities as unknowns, and the corrections applied.

1. Show that adding $\Delta$ to the clock $C$ of every satellite of system $g$ changes no position.
2. Show that adding a constant $d$ to one satellite's phase bias changes no position, and say what changes instead.
3. Show that adding $(\delta, \mu_2\delta)$ to the two code biases of one satellite is absorbed, and say what that does to the user's ionosphere.
4. The status report's clock comparison (step 5) removes two means before it computes a standard deviation. Which two, and which of the three cases does each stand for?

### Question 5.2 READ The header line of reference choices

*Builds on 4.2.* answer, p. 35

Line 2 of `runs/_rf0hv15_sim6/products.txt` is

```
S,S30000SIM,1,nominal-wu93,0,0,1,0,0.05,1,COD0MGXFIN_20241290000_01D_05M_ORB.SP3+
  COD0MGXFIN_20241290000_01D_30S_CLK.CLK,E:E02,G:G01,cbjump=1,atx=igs20.atx,oref=CoM,
  discfrac=0.25,hsig=pc,cbpin=E:C1C-C5Q,cbpin=G:C1C-C2W,
  sat_gf_datum=broadcast_tgd,pbhold=2
```

(one line in the file). With the writer `src/core/products/NetworkProductIO.cpp` (lines 465 to 602) and `include/products/NetworkProductEmitter.hpp`: what does each field declare? Which fields are the reference choices of Section 4? Which token does a current run add that this one lacks, and which choice does the real-data run's S line leave undeclared?

### Question 5.3 MEASURE A declared step, seen and absorbed

*Builds on 2.6 and 5.1.* answer, p. 36

Find the epochs at which the run declares a change of reference, then measure how every GPS clock record moves across the GPS one (tow 276150):

```
awk -F, '/^E,/ && $7=="1" {print NR": "$0}' runs/_rf0hv15_sim6/products.txt
C:/Python313/python.exe - <<'EOF'
import statistics as st
d, t = {}, None
for line in open('runs/_rf0hv15_sim6/products.txt'):
    f = line.rstrip('\n').split(',')
    if f[0] == 'E':
        t = round(float(f[2]) / 30) * 30
    elif f[0] == 'C' and f[1].startswith('G') and t in (276120, 276150):
        d.setdefault(f[1], {})[t] = float(f[2])
s = [v[276150] - v[276120] for v in d.values() if len(v) == 2]
print(len(s), 'satellites, mean step %.4f m, sd after the mean %.4f m'
      % (st.mean(s), st.pstdev(s)))
EOF
```

1. How large is the step of the GPS clock records, and how much of it is left after the per-system mean is removed?
2. Which move of Question 2.6 is this, and what absorbs it at a user?
3. Why would a grade of raw clock steps call this epoch a 1.6 m defect?

**Question 5.4** MEASURE **The reference satellite in every chord**

*Builds on 2.2 and 5.1.* answer, p. 37

The network's double-difference trace (`network_dd_fractional.csv`) differences every satellite against one reference satellite.

1. Derive: if every satellite's error has variance $\sigma^2$ and the reference satellite's $\sigma^2_{\text{ref}}$, what is the variance of one chord and the correlation between two chords?
2. Run (about 12 s) and compare the L1C error as traced and reference-free (median $|e|$ in cycles, and the share that rounds to the right integer), over all epochs and over the last quarter:

```
C:/Python313/python.exe scripts/simnet_amb_integer.py runs/simnet60_quiet_col_24h \
    --run runs/_rf0hv15_sim6 --ref-free
```

3. Why does a grade on the traced chords report the reference satellite's error?

**What this lesson establishes.**

- A float user absorbs a system-common clock change, a constant on a phase bias and a geometry-free code-bias offset; the reference choice is invisible to it.
- A declared step of 1.63 m on every GPS clock leaves 3 mm after the per-system mean: grade datum-invariant quantities, never raw steps.
- A double difference built on one reference satellite carries that satellite's error in every chord: grade reference-free. For the meeting: every number in the report is graded this way.

# 6 The satellite clock in time

**Goal.** See how the network ties each satellite clock to its past, what the phase-clock hold passes to a user, and why the clock's precision today is the a priori clock product's.

**The idea.** Three mechanisms shape the per-satellite part of the clock, the part a user cannot absorb.

1. *The temporal anchor* (card Q3, `network.clock_temporal_anchor`). Without it the clock is re-estimated from each epoch's data: the step between epochs was 9.23 cm per 30 s, because the clock's own process noise (`sat_clock_q`, $10^{-3}$ m$^2$/s) never binds. The anchor folds the previous epoch's clock in as a prior with variance $\max(\text{var} + q\,\Delta t,\ \text{floor})$ and `clock_anchor_q` $= 2.9 \times 10^{-8}$ m$^2$/s, a value measured from the clock's structure function (Figure 5, left).
2. *The exact coupling update* (card CTU, `carry_coupling_exact_update`). Carried station rows used to age the clock at $10^{-3}$ m$^2$/s while the shared solve folded it at $2.9 \times 10^{-8}$; the level froze. The exact update lets it converge at the anchor's rate (Figure 5, right).
3. *The phase-clock hold* (cards PBH and LAH, `emit_pb_hold_phase_clock`). Between declared resets the network keeps each satellite's clock minus phase bias, $C - P$, constant up to a common mode, so a user's phase rows follow the a priori clock.

Card SCB (24 October) made the consequence measurable on a simulated data set with injected satellite clock error (offset $\sigma$ 0.5 m, rate $\sigma$ $10^{-5}$ m/s, random walk $3 \times 10^{-8}$ m$^2$/s).

**Question 6.1** DERIVE **A spectral density as a step**

answer, p. 38

`clock_anchor_q` is $2.9 \times 10^{-8}$ m$^2$/s and the epoch 30 s.

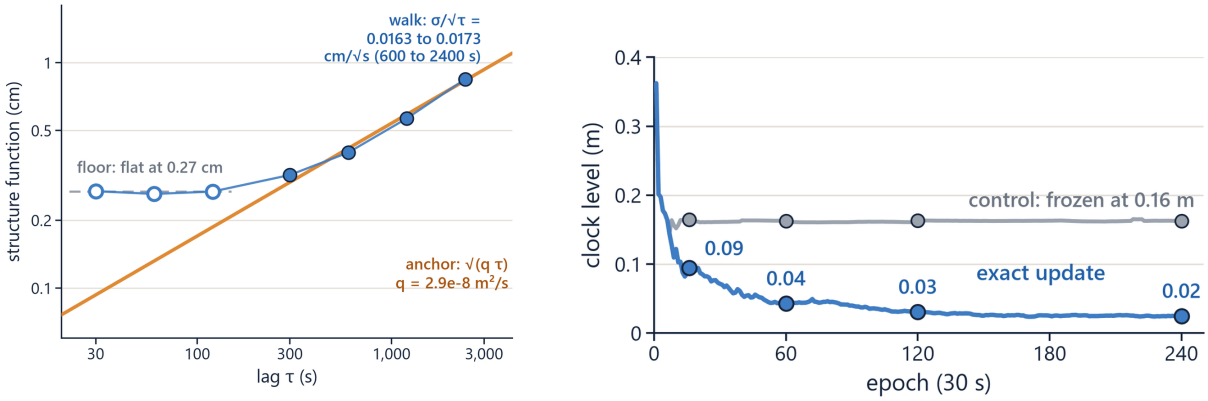



Figure 5: Left: structure function of the network's satellite clock correction against lag: a floor of 0.27 cm from 30 to 120 s, then growth along the anchor's line $\sqrt{q\tau}$, $q = 2.9 \times 10^{-8}$ m$^2$/s (deck slide 11). Right: simulated network, error of the clock level by epoch, graded from 00:30:15: the control freezes at 0.16 m, the exact coupling update falls to 0.02 m by epoch 240 (deck slide 10).

1. What is the $1\sigma$ random-walk step per epoch, and the $1\sigma$ accumulation over 6 h?
2. The injected rate on the clock data set has $\sigma$ $10^{-5}$ m/s, 0.30 mm per epoch, which card SCB calls "inside the golden anchor's permitted 0.93 mm per epoch". After how many epochs does a $10^{-5}$ m/s drift exceed the walk's $1\sigma$ accumulation? Why does comparing step sizes not guarantee that the anchor follows the drift?

### Question 6.2 DERIVE A spectral density as a time constant

*Builds on 6.1.* answer, p. 38

Treat the anchored clock as a scalar random walk observed once per epoch with noise variance $R = (9.23 \text{ cm})^2/2$ (the pre-anchor step is the difference of two independent epoch estimates). Use the steady-state gain $K = \sqrt{q\,\Delta t/R}$ of `docs/cards/raim-qc/DAY129-FRAME-ANATOMY.md` (lines 138 to 143).

1. What are $K$ and the time constant $\Delta t/K$?
2. What error does a $10^{-5}$ m/s drift leave? Compare with the coefficient card SCB measured for the reference configuration, 2127 s.
3. What does the same argument give for the self-contained regime's $q = 5 \times 10^{-6}$ m$^2$/s? Card SCB measured 52 s.

### Question 6.3 DERIVE What the phase-clock hold passes to the user

*Builds on 5.1 and 6.1.* answer, p. 39

`include/network/emit/PbHold.hpp` (lines 60 to 66) writes the held phase bias of a satellite as $P(t) = p(t_0) + [\tilde{c}(t) - \tilde{c}(t_0)]$, with $\tilde{c} = c - M$ the estimated clock minus its system mean.

1. Show what $C - P$ is between resets, and what a user's phase rows see when the a priori clock is $a(t)$ and the truth $a(t) + e(t)$.
2. On the clock data set $e(t)$ is a rate ($\sigma$ $10^{-5}$ m/s) plus a random walk ($3 \times 10^{-8}$ m$^2$/s). What rms error does the hold pass to the phase rows in hour 6 (5 to 6 h after the reset)? Card SCB, section 8, measured 0.201 m for GPS.

3. Lines 311 to 319 of `src/core/config/GnssConfigLoadNetwork.cpp` enforce a precondition on the hold. Which, and why does part 1 make it necessary? Which token of the S line declares the hold in force?

**Question 6.4** MEASURE **Two postures on the clock data set**

*Builds on 6.2 and 6.3.* answer, p. 40

Run the clock grader on the reference case and on the self-contained regime, both on the clock data set (a few seconds each):

```
C:/Python313/python.exe -B scripts/simnet_clock_truth_grade.py \
    --fixture D:/GnssProcessorSupport/fixtures/simnet60_clk_col_24h \
    --run D:/GnssProcessorSupport/runs_nom/_rap_scb_g_k6
C:/Python313/python.exe -B scripts/simnet_clock_truth_grade.py \
    --fixture D:/GnssProcessorSupport/fixtures/simnet60_clk_col_24h \
    --run D:/GnssProcessorSupport/runs_nom/_rap_scb_c_k6
```

Read for each the clock level at hour 6, the rms of the per-satellite mean errors, the correlation of the mean error with the drawn rate, and the user's hourly error in hours 3 to 6. Which posture follows the drift? Why is the reference case's user error so much larger than its clock error? What do these lines not show?

**Question 6.5** DERIVE **The cost of leaving the final clock, on real data**

*Builds on 6.4.* answer, p. 40

Card NBP, section 12 (lines 212 to 219), splits the cost of the native broadcast-clock posture over days 130 to 132 into three steps: the self-contained regime +0.0024 m, removing satellite G01 −0.0007 m, the broadcast clock in place of CODE's +0.0099 m. The pooled median of the whole change is +0.0171 m (line 176).

1. Why do the steps not add to the total? Which part is the clock's information?
2. Against which reference configuration, which days and which hours was the 1.7 cm measured, and what does the first hour add?

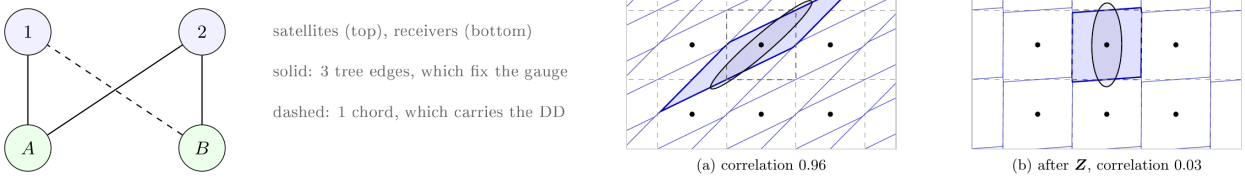
**What this lesson establishes.**

- $q = 2.9 \times 10^{-8}$ m$^2$/s is a 0.93 mm step per epoch and, against the epoch noise, a time constant of about 35 min.
- The phase-clock hold hands a user the a priori clock's motion: a wrong a priori clock reaches the phase rows one to one, so the hold is valid only with a precise a priori clock.
- The self-contained postures estimate the clock to about 2 cm on simulated error but are not armed; on real data, replacing CODE's clock costs 1.7 cm at the median. For the meeting: the clock's precision is CODE's.

# 7 Float ambiguities and integers

**Goal.** See why the network keeps its ambiguities float, what a complete set of integers would buy, and why the corrections carry no integers yet.

**The idea.** In the uncombined model every phase row carries both the slant ionosphere and an ambiguity, and on GPS L1 and L2 the two enter in nearly the same direction: a common L1 and L2 cycle moves the phase by $(\lambda_1, \lambda_2) = (0.1903, 0.2442)$ m, the ionosphere along $(1, \mu_2)$ with $\mu_2 = 1.6469$. With a free slant ionosphere per line of sight, the integers become visible only once the double-differenced ionosphere is known to about 0.1 L1 cycle. The re-founding chose the *float vehicle* (card RF-1): each ambiguity is a float state of its station block with a joint covariance, a floored birth prior and zero process noise; integers are graded, not fixed. Question 2.5 showed where a user's integer property comes from: the network's pins.




(a) correlation 0.96 (b) after $\boldsymbol{Z}$, correlation 0.03

Figure 6: Left: the smallest observation graph with a cycle: satellites 1 and 2 (top), receivers $A$ and $B$ (bottom); the three solid arcs form a spanning tree whose ambiguities the reference choice fixes, the dashed chord carries the one double difference the data determine (`docs/report/network_carry.tex`, figure 1). Right: a two-dimensional integer search; at correlation 0.96 the error ellipse is long and the search's pull-in region a long parallelogram, unlike the rounding squares (dashed); after the decorrelating transformation $\boldsymbol{Z}$ the two nearly agree (`docs/report/linear_algebra.tex`, figure 3; a generic example).

### Question 7.1 DERIVE **Why an integer is mostly ionosphere**

*Builds on 1.2 and 1.3.* answer, p. 41

From $f_1 = 1575.42$ MHz and $f_2 = 1227.60$ MHz, compute $\lambda_1$, $\lambda_2$ and $\mu_2$, the angle between the ionosphere direction $(1, \mu_2)$ and the common-cycle direction $(\lambda_1, \lambda_2)$ in metres of phase, its cosine, and the part of one common L1 and L2 cycle that is orthogonal to the ionosphere. Card ARL (line 43) calls a narrow-lane integer "99.3 % ionosphere". Is that a share of amplitude or of variance?

### Question 7.2 DERIVE **An ionosphere error in cycles**

*Builds on 7.1.* answer, p. 41

A double-difference L1 ionosphere error $d$ (metres) is absorbed by the ambiguities.

1. What are the L1 and wide-lane ($L1 - L2$) errors in cycles for $d = 0.4233$ m (no pin) and $d = 0.0419$ m (every ambiguity pinned)? Check the factors against `docs/cards/raim-qc/T3-IONO-MECHANISM.md` (lines 326 to 331).
2. Instantaneous resolution needs about 0.1 L1 cycle. What is that in metres and in TECU, and by what factor does the no-pin case miss it?

### Question 7.3 DERIVE **Integer pins and float pins**

*Builds on 2.5 and 4.2.* answer, p. 42

Work Question 2.5 in cycles, with a receiver phase bias $\beta_r$ (the same for every satellite), so that $N_r^i = z_r^i + \beta_r$ with $z_r^i$ an integer. Let the network pin $N_A^i \equiv \zeta^i$ instead of 0.

1. What does the network emit as $\bar{b}^i$, and what does the user $C$ estimate as $\tilde{N}_C^i$?

2. Form the difference between satellites $i$ and $k$ at $C$. When is it an integer?
3. The reference configuration pins each anchor ambiguity at its own float value (`docs/network_ambiguity_resolution_design.md`, line 335). What does part 2 predict for its corrections? What does card IRD (sections 1 to 3) propose instead?

**Question 7.4** READ **Complete and partial pins**

*Builds on 7.2.* answer, p. 42

Read sections 0 and 8.1 of `docs/SPEC_oracle_pin.md`. What is the "integer-information deficit", which row of its table carries it, and why is the raw per-line-of-sight error (2.73 m) the wrong row? What do the partial pins (L5Q alone; L1C with L2W; one station; one satellite) show, and on which simulated data set were the numbers produced?

**Question 7.5** MEASURE **Integers in the float, none in the corrections**

*Builds on 7.3.* answer, p. 43

Run (each under 10 s):

```
C:/Python313/python.exe scripts/simnet_wire_integer_consistency.py \
    runs/simnet60_quiet_col_24h --run runs/_rf0hv15_sim6 --bands L1C,L2W \
    --systems G --wl --nl --null-mc 500 --start-hms 00:30:15
C:/Python313/python.exe scripts/simnet_producer_ar_grade.py runs/simarl_r2 \
    --run runs/_rap_arl_r2dp --settle-h 1
```

The first grades the published GPS phase biases for integer consistency (the fraction of wide-lane and narrow-lane combinations against a uniform null; a set of perfect corrections reads 0.000). The second grades the log-only integer search of card ARL. What does each print, and what do the two together say about where the integers are?

**What this lesson establishes.**

- An L1 and L2 integer step lies 6.66° from the ionosphere direction; with a free slant ionosphere the integers need a double-differenced ionosphere to 0.117 TECU.
- A user's between-satellite ambiguity is an integer only if the network's pins are integers; the reference configuration pins float values, so its corrections sit at the uniform null.
- The network's float solution holds integers (a log-only search fixes 42 % on a simplified simulation, none wrongly). For the meeting: no integers are published or fixed on real data.

# 8 Testing on simulated data

**Goal.** See what the simulated network certifies, where a simulated run starts and is graded, and how much error the network adds above the user's own floor.

**The idea.** The simulated data set `runs/simnet60_quiet_col_24h` holds 60 network stations and the test station SZZ000SIM at 30 s from 00:00:30 (Figure 7, left). It is *quiet*: no cycle slips and no satellite clock error, because its truth clock is the CODE product that the network also takes as its a priori clock. The simulator extrapolated the satellite positions outside its SP3 window for its first epochs; the reference case is graded from 00:30:15. The network reports clocks and biases in its own reference choices, so truth is moved into that basis before any comparison. The simulator also writes a set of perfect corrections, so the same test station can be positioned with both sets (Figure 7, right).

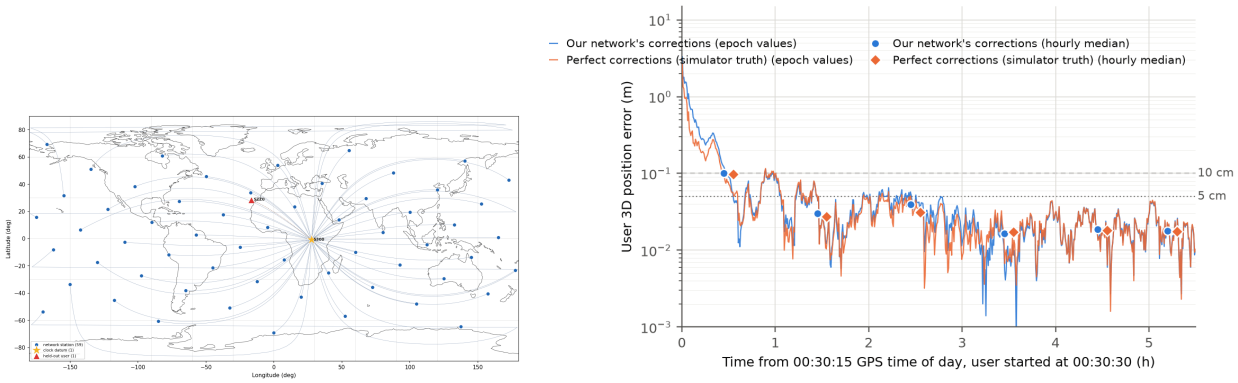



Figure 7: Left: the simulated network, 59 stations and the clock-datum station S30000SIM (star); the test station SZZ000SIM (triangle) is left out (deck slide 15). Right: SZZ000SIM with our network's corrections (blue) and with perfect corrections (orange), user started at 00:30:30; lines are epochs, markers hourly medians (`results/t2/plain/t2_sim_floor_restart.png`).

### Question 8.1 READ What the quiet simulation cannot test

*Builds on 6.3.* answer, p. 43

Read card SCB, section 1 and the start of section 2 (lines 17 to 25), and `results/t2/results.md` (lines 123 to 126). What makes the clock untestable on the quiet data set, and what does its 0.9 cm agreement certify instead? Check the absence of injected clock error directly:

```
ls runs/simnet60_quiet_col_24h/*/truth_satclock.csv 2>/dev/null | wc -l
ls D:/GnssProcessorSupport/fixtures/simnet60_clk_col_24h/*/truth_satclock.csv | wc -l
```

### Question 8.2 DERIVE Where a simulated run starts

answer, p. 43

1. The data set's first epoch is tow 259230 (00:00:30). An epoch is reached when its elapsed time since the first epoch is at least the offset (`include/math/TimeUtils.hpp`, lines 416 to 448). At which tow and time do `network.start_offset_hours` 0.504167 and 0.4958333333 start a run? What does the "half-epoch back-off" do?
2. Count the extrapolated epochs. The geometry log warns only when the extrapolation exceeds half the SP3 interval (`src/core/model/ModelGeometry.cpp`, line 1204):

   ```
   grep "Position extrapolation" \
       runs/simnet60_quiet_col_24h/SZZ000SIM/logs/geometry.log \
     | sed -E 's/.*tow ([0-9.]+):.*/\1/' \
     | awk '{c[int(($1+15)/30)*30]++} END{for(e in c) print e,c[e]}' | sort -n
   ```

   How many epochs are logged, how many precede the SP3 window (which opens at tow 259500), and is "faulty in the first 30 minutes" the right description?

### Question 8.3 MEASURE The published baseline's clock in the network's basis

*Builds on 4.4, 5.1 and 8.2.* answer, p. 44

Run the clock grader on the published simulated baseline (started 00:00:30) and on card SCB's quiet case (started 00:31:00):

```
C:/Python313/python.exe -B scripts/simnet_clock_truth_grade.py \
    --fixture runs/simnet60_quiet_col_24h --run runs/_rf0hv15_sim6 --hours 6
C:/Python313/python.exe -B scripts/simnet_clock_truth_grade.py \
    --fixture runs/simnet60_quiet_col_24h \
    --run D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6
```

Compare the clock level by hour and the rms of the per-satellite mean errors. Rerun the first with `--no-ds`: what does the datum share remove, and why is it added to the truth rather than subtracted from the estimate? Why does the first run's user not show its per-satellite constants?

**Question 8.4** DERIVE **The network's cost and the user's floor**

*Builds on 8.2.* answer, p. 44

From `results/t2/results.md` (lines 195 to 206), the hourly 3D medians with the user restarted at 00:30:30, in metres: perfect corrections 0.0983, 0.0275, 0.0311, 0.0173, 0.0181, 0.0176; our network 0.1013, 0.0301, 0.0396, 0.0166, 0.0186, 0.0178. Compute the network's cost per hour, the largest cost and the floor in hours 4 to 6. In which component does the floor sit (`results/t2/metrics.csv`)?

**What this lesson establishes.**

- The quiet simulation has no clock error to estimate; it certifies the algebra and the biases.
- Nine epochs are extrapolated; 00:30:15 is a grading origin, and a run started at 00:00:30 keeps per-satellite clock constants of 0.32 m that no float user can see.
- With perfect corrections as the control, the network adds at most 0.9 cm; the 1.7 to 1.8 cm floor is the user side's, in the height. For the meeting: one realization, no slips, no clock error.

# 9 Real data, and what may be claimed

**Goal.** Grade the real-data result the way the project does, and put the course together into what the presentation may claim.

**The idea.** The real-data reference case (`config/golden/golden_field_refound.json`, run `runs/_rf0hv15_field6`) processes 58 IGS stations on 8 May 2024, GPS L1/L2 and Galileo E1/E5a. The test station WTZZ00DEU is left out of the network and positioned as a moving receiver from the clock and bias records, without integer fixing and with its own float ionosphere; it shares its site with two network stations. Sixteen further stations, none of them in the network, are positioned from the same file afterwards. A change is graded against the baseline pair and an unchanged copy run by the same software, on the test station by hour and on the 16 stations, never on one alone; the first hour is excluded from settled claims (`CLAUDE.md`, "Network runs: start from a golden").

**Question 9.1** MEASURE **The test station and the 16 stations**

*Builds on 5.4.* answer, p. 45

Run (about 2 s and 14 s):

```
C:/Python313/python.exe scripts/rapid_arm.py score \
    runs/_rf0hv15_field6 runs/_rf0hv15_sim6
C:/Python313/python.exe scripts/clock_level/gcensus.py grade galpc galp galpg
```

Read the test station's median error by hour on both data sets, the 16 stations' median by hour and the paired Galileo result. By what factor is hour 1 worse than the mean of hours 2

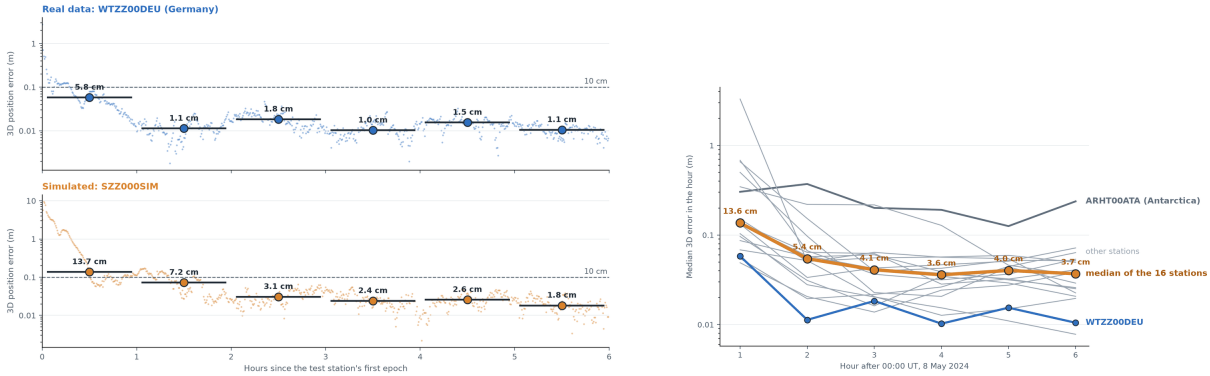



Figure 8: Left: the test station with our network's corrections, real data (WTZZ00DEU, top) and simulated (SZZ000SIM, bottom); dots are epochs, bars hourly medians. Right: hourly median for the 16 further stations (grey), their median (orange), WTZZ00DEU (blue) and ARHT00ATA (dark). Status report figures.

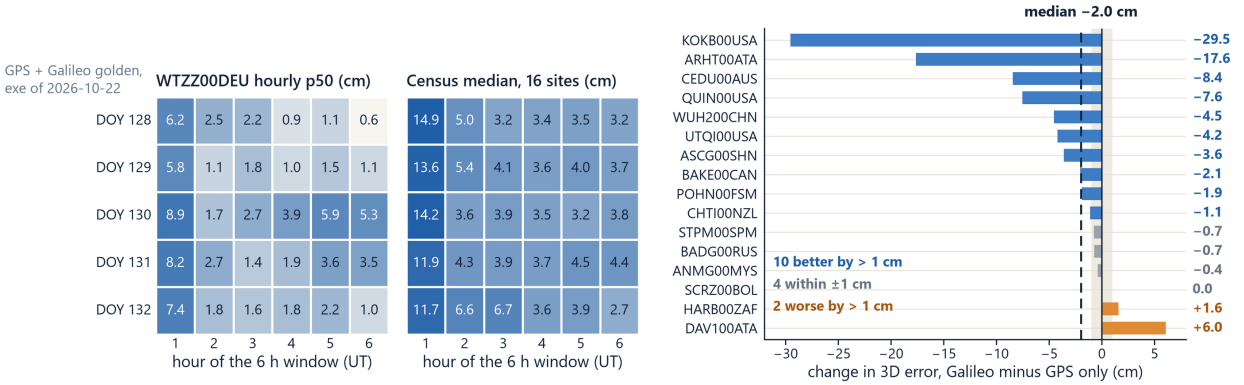



Figure 9: Left: five days, 7 to 11 May 2024, hourly median in cm: WTZZ00DEU and the median of the 16 stations (the labels "golden" and "census" are the record's names for the reference configuration and the 16 stations; deck slide 33). Right: change of each station's error in hours 2 to 6 when Galileo is added, median −2.0 cm (deck slide 19).

to 6, at WTZZ00DEU and at the median station? In which mode is WTZZ00DEU positioned (`runs/_rf0hv15_field6/arm.json`: `filter.process_noise.position_q`, `network.closed_loop.ar_mode`), and which network stations share its site?

### Question 9.2 DERIVE Five days from the stored grades

*Builds on 9.1.* answer, p. 46

`D:/GnssProcessorSupport/runs_nom/mday_galileo_grades.txt` holds, per day 128 to 132, the 16 stations' median by hour and WTZZ00DEU by hour. From hours 2 to 6 of the five days, compute the median and the mean of the 25 sixteen-station medians and the median of the 25 WTZZ00DEU values. On which day is WTZZ00DEU worse than the median station in hours 4 to 6, and what does that do to "the best station"?

### Question 9.3 READ The clock against CODE and GFZ

*Builds on 5.1 and 6.3.* answer, p. 46

Read `results/t3/table_clock_report.tex` and `results/t3/results.md` (the summary table

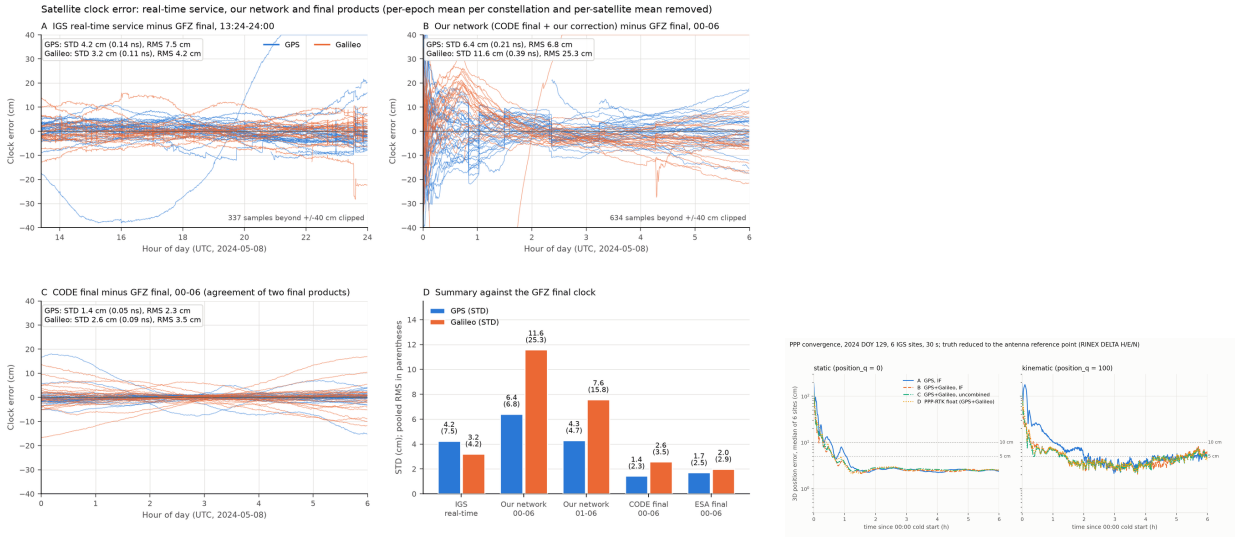



Figure 10: Left: satellite clock error against the GFZ final clock after removing the per-system epoch mean and each satellite's mean: (A) IGS real-time service, (B) our network, CODE final plus our correction, 00:00 to 06:00 with the first hour, (C) CODE final, (D) summary (`results/t3/fig_clock_limits_plain.png`). Right: the PPP comparison of steps 1 to 3, median over six stations, static and moving receiver (`results/t1/fig_t1_convergence.png`).

and lines 100 to 113). Which clock is compared with which, over which hours? Why is GFZ the reference and not CODE? Over the same hours, how does the network's corrected clock compare with CODE's alone? What do the correlations of the correction with (GFZ minus CODE) and (ESA minus CODE) say?

### Question 9.4 SYNTHESIS **What the presentation may claim**

*Builds on the whole course.* answer, p. 47

For each of five topics, state in two or three sentences what the presentation of 29 October may claim and the evidence that bounds it: (a) the model and its per-epoch check; (b) the reference choices; (c) the satellite clock; (d) integer ambiguities; (e) the simulated and real-data results.

**What this lesson establishes.**

- On 8 May 2024 the test station reaches 1.0 to 1.8 cm and the median of the 16 further stations 3.6 to 5.4 cm in hours 2 to 6, without integers.
- Over five days the median station holds 3.8 cm; the test station is the best station on some days, not all, and shares its site with network stations.
- The network's corrected clock is noisier than CODE's alone and carries no clock information: the user's precision is CODE's.

## Answers

The key is written to teach. Each answer opens with a *Strategy*: the principle to apply and the first move, enough to restart a stalled attempt without giving the result away. The *Solution* then works the question step by step and says why each step is taken. *Why it works* names the idea underneath, *Pitfall* the mistake most often made with it, and *Takeaway* the result that later questions use. Each answer gives its source: the card and lines, the file and lines, or the run. Where a command reproduces the answer, *Check* gives it. Answers marked *checked* were reproduced on 24 October 2026 (the list is in the last section). Cards named by a short name are files in `docs/refound/`: ALD `ALD_anchor_level_datum_2026-10-23.md`, ARL `ARL_integer_ambiguity_ladder_2026-10-09.md`, CTU `CTU_coupling_time_update_2026-09-28.md`, FCFC `FCFC_fixed_cycle_ledger_2026-10-11.md`, GAD `GAD_galileo_adoption_2026-10-13.md`, IRD `IRD_integer_recovering_datum_2026-10-13.md`, LAH `LAH_level_aware_hold_2026-09-29.md`, NBP `NBP_native_broadcast_clock_posture_2026-10-07.md`, R1D `R1D_station_level_datum_2026-10-23.md`, SCB `SCB_sim_clock_error_bed_2026-10-24.md`, SCL `SCL_self_contained_clock_level_2026-10-03.md`. The S-system cards `SBASIS-THEORY-E.md` and `SBASIS-CONTRACT.md` are in `docs/cards/network-vehicle/`. "The worked example" is `docs/report/network_example.tex` (Mathematical Reference, "Worked example: estimability in a two-receiver network"). Status report sections are cited by name; its wording was qualified on 24 October with the boundaries of Question 9.4.

### Lesson 1: one receiver

**Answer 1.1** (question, p. 3)

**Strategy.** A null direction is a change of the unknowns that cancels inside every row. Look for unknowns that always appear together, with fixed coefficients, and move them in opposite senses.

**Solution.**

1. Every code row contains the pair $c\,\mathrm{d}t_r + B_{r,s}$ and every phase row the pair $c\,\mathrm{d}t_r + b_{r,s}$. Raising the clock by $\delta$ and lowering all four biases by $\delta$ changes each pair by $\delta - \delta = 0$, for every satellite. No observation moves, so the data cannot tell the two solutions apart: one null direction per receiver.
2. Write the two code biases as $B_{r,1} = B_{r,\mathrm{IF}} + d_r$ and $B_{r,2} = B_{r,\mathrm{IF}} + \mu_2 d_r$. Subtracting the first from the second gives $d_r = (B_{r,2} - B_{r,1})/(\mu_2 - 1)$, a scaled receiver DCB. Substituting back gives $B_{r,\mathrm{IF}} = (\mu_2 B_{r,1} - B_{r,2})/(\mu_2 - 1)$, and with $\mu_2 = f_1^2/f_2^2$ this is $(f_1^2 B_{r,1} - f_2^2 B_{r,2})/(f_1^2 - f_2^2)$: the weights of the ionosphere-free combination.
3. The code row becomes $\cdots + (c\,\mathrm{d}t_r + B_{r,\mathrm{IF}}) + \mu_s(I_r^i + d_r) + \ldots$. The clock never appears without $B_{r,\mathrm{IF}}$, so the estimated "receiver clock" is the ionosphere-free clock $c\,\mathrm{d}t_r + B_{r,\mathrm{IF}}$.

**Why it works.** The split sorts a bias by how it behaves across frequencies. The part that is the same on both frequencies behaves exactly like a clock and merges with it; the part that scales with $\mu_s$ behaves exactly like the ionosphere (Question 1.3). Whatever enters every row identically cannot be separated from the clock.

**Pitfall.** Treating an estimated receiver clock as the physical clock, or comparing receiver clocks from two programs that split the biases differently. Two correct programs can disagree by $B_{r,\mathrm{IF}}$.

**Takeaway.** The satellite side mirrors this: the network's satellite clock is the ionosphere-free clock, clock plus ionosphere-free code bias (Question 4.4).

*Source.* status report, "One observation model"; `SBASIS-THEORY-E.md` lines 87 to 92 (the equations) and 158 to 160 (the ionosphere-free clock).

**Answer 1.2** (question, p. 4)

**Strategy.** Repeat the test of Question 1.1 between the receiver phase bias and the ambiguities. The ambiguity is in cycles, so a shift of $\delta_s$ metres in the bias is a shift of $\delta_s/\lambda_s$ cycles in the ambiguity.

**Solution.**

1. $\lambda_s(N^i_{r,s} + \delta_s/\lambda_s) + (b_{r,s} - \delta_s) = \lambda_s N^i_{r,s} + b_{r,s}$, for every satellite at once. Each signal has its own $\delta_s$, so a dual-frequency receiver has two such directions (type 4 of the network table counts $f(n-1)$ of them, Question 2.4).
2. With no receiver phase-bias state, the float ambiguity takes the bias in: it estimates $N^i_{r,s} + b_{r,s}/\lambda_s$ (and, without satellite phase biases, $-b^i_s/\lambda_s$ as well). The receiver phase bias is an arbitrary fraction of a cycle, the same for all satellites, so the float value is no integer however exact the satellite biases are.
3. The difference between satellites $i$ and $k$ cancels $b_{r,s}$, which both carry. What is left is $N^i - N^k$ plus the difference of the satellite biases, so it is an integer only if the product supplies those biases exactly (the observable-specific biases of the report's step 3).

**Why it works.** Any constant in all phase rows of one signal at one receiver can be moved into that signal's ambiguities, because each ambiguity is itself a constant in its row. Only a difference that cancels the constant can carry integer meaning.

**Pitfall.** Rounding a PPP float ambiguity, or expecting the product's satellite phase biases alone to restore integers: the receiver's part stays until a difference between satellites removes it.

**Takeaway.** This is why "a constant satellite phase bias simply joins the ambiguity, so case D matches case C" (report, step 3), and why integers appear only in differences (Questions 2.3 and 7.3).

*Source.* `SBASIS-THEORY-E.md` lines 97 to 104 (type 4); status report, step 3.

**Answer 1.3** (question, p. 4)

**Strategy.** Use the split of Question 1.1 and look at what multiplies $\mu_s$ in the code rows: any two unknowns with the same coefficient pattern can trade value.

**Solution.**

1. $\mu_s I^i_r + B_{r,s} = B_{r,\mathrm{IF}} + \mu_s(I^i_r + d_r)$: on code the ionosphere and $d_r$ always appear as one sum.
2. Raising every slant ionosphere by $\delta$ and lowering $d_r$ by $\delta$ leaves each code row as it was. On phase the ionosphere enters with the opposite sign, so every phase row gains $-\mu_s\delta$. That is a constant per signal, common to all satellites, so the float ambiguities absorb it ($N^i_{r,s} \to N^i_{r,s} + \mu_s\delta/\lambda_s$). One more null direction per receiver.
3. The estimated slant ionosphere is $I^i_r + d_r$ (and, with a product's satellite terms, minus that product's satellite geometry-free bias). It becomes the physical ionosphere only once $d_r$ is fixed by a convention, such as a zero-mean condition on the receivers' DCBs, or separated by a vertical ionosphere model.

**Why it works.** On code the ionosphere and the geometry-free bias have the same signature, $\mu_s$; on phase the ionosphere's opposite sign looks like a per-signal constant, which float ambiguities soak up. This is Odijk's $\tilde{\iota} = \iota + d_{r,\mathrm{GF}} - d^s_{\mathrm{GF}}$ (types 2b and 3b).

**Pitfall.** Comparing a user's or the network's slant ionosphere with a global map without removing the receiver and satellite geometry-free biases: the difference is a convention, not an ionospheric error.

**Takeaway.** The network's two conventions for this direction are the per-satellite geometry-free pin at the broadcast $T_{GD}$ and the zero-mean receiver DCB prior (Section 4).

*Source.* `SBASIS-THEORY-E.md` lines 122 to 128.

## Lesson 2: two receivers

**Answer 2.1** (question, p. 5)

**Strategy.** Each code row has $+1$ in its receiver's column and $-1$ in its satellite's. An entry $N_{jk}$ of $\boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ sums, over the rows, the product of the coefficients of unknowns $j$ and $k$: count co-occurrences and their signs.

**Solution.**

1. A receiver clock is in four rows ($N_{jj} = 4$); a satellite clock in two ($N_{jj} = 2$); each receiver and satellite share exactly one row with signs $+1$ and $-1$ ($-1$); two receivers, or two satellites, never share a row (0):
$$\boldsymbol{N} = \begin{bmatrix} 4\boldsymbol{I}_2 & -\boldsymbol{J}_{2\times 4} \\ -\boldsymbol{J}_{4\times 2} & 2\boldsymbol{I}_4 \end{bmatrix}, \qquad \boldsymbol{J} \text{ all ones.}$$
2. Each row of $\boldsymbol{N}$ sums to zero ($4 - 4 = 0$ and $-1 - 1 + 2 = 0$), which is the statement $\boldsymbol{N}\mathbf{1} = \mathbf{0}$. With one independent null vector the rank is $6 - 1 = 5$.
3. $\mathbf{1}$ adds the same $\kappa$ to all six clocks. Each observation is a difference $a_r - g^i$, so none changes. More epochs or satellites only add rows of the same form, so the common level stays undetermined: one reference choice is needed. *Checked.*

**Why it works.** A row sum of zero is the algebraic form of "only differences are observed". Every row of $\boldsymbol{A}$ annihilates $\mathbf{1}$, hence so does $\boldsymbol{N}$.

**Pitfall.** Believing that more data cures a rank deficiency. Data of the same kind add rows in the same space; only a different kind of information (a pin, a prior, an outside clock) fixes the level.

**Takeaway.** This is type 1a. In the real network it survives the elimination of the stations as the clock direction the rank gate counts (Questions 3.1 and 3.5).

*Source.* the worked example, eq. ex-code-normals.

*Check.*

```
C:/Python313/python.exe - <<'EOF'
import numpy as np
A = np.array([[1 if j == r else 0 for j in range(2)]
              + [-1 if k == i else 0 for k in range(4)]
              for r in range(2) for i in range(4)])
N = A.T @ A
print(N, np.linalg.matrix_rank(N), N @ np.ones(6))
EOF
```

**Answer 2.2** (question, p. 6)

**Strategy.** With one clock pinned, every observation $P_r^i = a_r - g^i$ has one unknown left as soon as its neighbour is known: walk through the table from the pinned clock.

**Solution.**

1. Pin $a_A \equiv 0$: receiver $A$'s rows give $g^i = -P_A^i = (-3, -6, -1, -4.5)$; satellite 1 then gives $a_B = P_B^1 + g^1 = -2$. Pin $g^1 \equiv 0$: $a_A = P_A^1 = 3$ and $a_B = P_B^1 = 1$; then $g^i = a_A - P_A^i = (0, -3, 2, -1.5)$.
2. The second solution minus the first is $+3$ on every unknown: $3 \cdot \mathbf{1}$. Both solutions satisfy $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{y}$, so their difference satisfies $\boldsymbol{A}(\boldsymbol{x} - \boldsymbol{x}') = \mathbf{0}$ and must lie in the null space.
3. $a_A - a_B = 2$, $g^2 - g^1 = -3$, $g^3 - g^1 = 2$ and $a_B - g^4 = 2.5$ in both solutions and in the truth. Each combination has coefficients $\boldsymbol{f}$ with $\boldsymbol{f}^{\mathrm{T}}\mathbf{1} = 0$, so adding $\kappa\mathbf{1}$ cannot change it.
4. $\hat{g}^2 = g^2 - a_A = -1 - 5 = -6$: the true clock shifted by the reference receiver's clock. *Checked.*

**Why it works.** A combination is estimable exactly when it is blind to the null space. The reference choice selects one point on the line $\boldsymbol{x} + \kappa\mathbf{1}$; an estimable combination has the same value at every point of that line.

**Pitfall.** Comparing satellite clocks from two solutions with different references (two networks, or two days with different pivots) and reading the offset as an error. The offset is the reference choice.

**Takeaway.** This is the datum-invariance theorem in six unknowns (Question 4.1), and the reason a network declares its pivots (Question 5.2).

*Source.* the worked example, tables "Two datum choices, same fit" and "What is invariant".

**Answer 2.3** (question, p. 6)

**Strategy.** Count first: unknowns, observations, null directions, rank. Then use the reference choice to read the phase unknowns off the table one satellite at a time, using $a_A - g^i = P_A^i$ from the code.

**Solution.**

1. Unknowns $6 + 4 + 8 = 18$; observations 16. Null directions: the clock direction of Question 2.1, plus one per satellite, $(b^i + c_i,\ N_A^i - c_i,\ N_B^i - c_i)$, which changes no phase row: 5 in all, rank 13, redundancy $16 - 13 = 3$. The redundancy is the three code double differences, $(n-1)(m-1)$, each identically zero here.
2. With $A$ alone, each satellite's phase is one equation in $b^i + N_A^i$: the pair cannot be separated and there is no redundancy. With $B$ there are two equations in three unknowns, rank 2: the new estimable quantity is $N_A^i - N_B^i$, the between-receiver single difference.
3. With $a_A \equiv 0$ and $N_A^i \equiv 0$: $\bar{b}^i = \Phi_A^i - P_A^i = (10.10, -5.20, 20.05, 8.15)$, then $\hat{N}_B^i = \Phi_B^i - P_B^i - \bar{b}^i = (2, 2, -2, 1)$.
4. $\bar{b}^1 = b^1 + N_A^1 = 0.10 + 10$: the emitted bias carries the pinned receiver's ambiguity, Odijk's $\bar{b}^i = b^i +$ (pivot-receiver ambiguity). $\hat{N}_B^i = N_B^i - N_A^i$, the single difference an RTK user fixes. *Checked.*

**Why it works.** A satellite phase bias is shared by all receivers, an ambiguity by one. Their only common freedom is the per-satellite shift; pinning one receiver's ambiguity per satellite removes it and leaves differences between receivers.

**Pitfall.** Reading 10.10 m as a physical, implausibly large hardware bias. It is a lump: bias plus the pinned ambiguity.

**Takeaway.** What a user's ambiguities become depends on these pins: integer pins give integer differences, float pins do not (Questions 2.5 and 7.3).

*Source.* the worked example, "Adding carrier phase".

**Answer 2.4** (question, p. 6)

**Strategy.** Count the null directions two ways and check that they agree: by the type table, and by unknowns minus rank.

**Solution.**

1. $1+2+1+4+1+4=13$, Eq. 24 with $f=1$, $n=2$, $m=4$: $1+2+2(1+4)=13$.
2. The full model has 26 unknowns (clocks 6; code biases $2+4$; receiver phase biases 2; satellite phase biases 4; ambiguities 8). The lumped model has 18: the lumps removed $B_r$, $B^i$ and $b_r$, 8 unknowns. Each removed column was a combination of columns that stay (a code-bias column is a clock column minus phase columns), so the rank stays 13 and the null space shrinks from $26-13=13$ to $18-13=5$. Gone: types 1b (2), 2a (1), 3a (4) and 4 (1). Left: 1a (1) and 5 (4), the five of Question 2.3. *Checked.*

**Why it works.** Removing an unknown whose column lies in the span of the others removes one null direction and no information: the data determine exactly what they did before. A lump is therefore a reference choice made by definition.

**Pitfall.** Fearing that lumping loses information. The rank, which is the information, is unchanged; only the bookkeeping of what each number means changes.

**Takeaway.** The project lumps in the same way (the receiver phase bias has no state, Question 3.4); the accounting from Eq. 24's 465 to the gate's 2 (Question 3.5) is this question at scale.

*Source.* `SBASIS-THEORY-E.md` lines 97 to 114 (the table and Eq. 24).

**Answer 2.5** (question, p. 6)

**Strategy.** Write $C$'s equations with the truth, substitute the corrections the network emits, and collect what is left on each unknown.

**Solution.**

1. $P_C^i = a_C - g^i = (2,5,0,3.5)$; $\Phi_C^i = a_C - g^i + b^i + N_C^i = (13.10, 0.80, 19.05, 13.65)$.
2. Code: $P_C^i = \hat{a}_C - \hat{g}^i$ gives $\hat{a}_C = P_C^i + \hat{g}^i = -1$ for every satellite. Phase: $\tilde{N}_C^i = \Phi_C^i - \hat{a}_C + \hat{g}^i - \bar{b}^i = (1,1,-1,2)$.
3. Since $\hat{g}^i = g^i - a_A$, $C$'s clock absorbs $-a_A$: $\hat{a}_C = a_C - a_A = 4-5$. Since $\bar{b}^i = b^i + N_A^i$, its ambiguities absorb $-N_A^i$: $\tilde{N}_C^i = N_C^i - N_A^i$. $C$'s solution carries receiver $A$'s clock and ambiguities, and $C$ never needs to know $A$: "the user never needs to know the network pivot". *Checked.*

**Why it works.** The offsets the network's reference puts into the corrections are exactly the kinds a user's own unknowns can absorb: one common to all satellites (the clock) and one per satellite (the ambiguity).

**Pitfall.** Grading a user's clock or float ambiguities against truth: both are relative to the network's reference.

**Takeaway.** PPP-RTK is relative positioning. Section 5 generalizes the two absorptions; Question 7.3 asks when $\tilde{N}_C^i$ is an integer.

*Source.* `SBASIS-THEORY-E.md` lines 166 to 170.

**Answer 2.6** (question, p. 7)

**Strategy.** Every admissible solution is the pinned solution plus some $\kappa\mathbf{1}$. Each reference rule fixes $\kappa$; compute it before and after the change.

**Solution.**

1. Under $g^1 \equiv 0$, $\sum g^i = -2.5$. Sum-to-zero needs $\kappa = 2.5/4 = 0.625$: $g = (0.625, -2.375, 2.625, -0.875)$ and $a = (3.625, 1.625)$.
2. Pin: with satellites 1 to 3, $g = (0, -3, 2)$ and $a = (3, 1)$; nothing moves. Sum-to-zero: $\sum g^i = -1$ over three satellites, $\kappa = 1/3$, so $g = (0.333, -2.667, 2.333)$ and $a = (3.333, 1.333)$: every clock moves by $1/3 - 0.625 = -0.292$ m.
3. The pin moves to satellite 2, whose clock was $-3$: $\kappa = +3$, so every clock moves by $+3$ m ($g = (\cdot, 0, 5, 1.5)$, $a = (6, 4)$).
4. No estimable quantity moves: every change is along $\mathbf{1}$. A user absorbs the common step in its receiver clock at that epoch, but anything it carries across the epoch at the old level sees a step. The epoch must be declared, and the user resets across it. *Checked.*

**Why it works.** Sum-to-zero defines $\kappa$ from the current satellite set, so it changes at every rise and set ("steps at every constellation-membership change"); a pin defines $\kappa$ from one satellite, so it changes only when that satellite leaves, a handover the user "pays for with an ambiguity reset".

**Pitfall.** Reading a declared reference step in the corrections as a defect; it vanishes from every estimable quantity (Question 5.3).

**Takeaway.** The project pins one clock per constellation with a sticky pivot and declares each handover; Question 5.3 measures one in a real run.

*Source.* `SBASIS-THEORY-E.md` line 145; `include/network/solve/Sbasis.hpp` lines 162 to 166.

## Lesson 3: the real network

**Answer 3.1** (question, p. 8)

**Strategy.** Partition $\boldsymbol{N}$ into receiver and satellite blocks. The receiver block is $4\boldsymbol{I}$, trivial to invert; the rest is matrix products with $\boldsymbol{J}$, whose square is a multiple of itself.

**Solution.**

1. $\boldsymbol{N}_{ga}\boldsymbol{N}_{aa}^{-1}\boldsymbol{N}_{ag} = \frac{1}{4}(\boldsymbol{J}_{4\times2}\boldsymbol{J}_{2\times4}) = \frac{2}{4}\boldsymbol{J}_4$, so $\bar{\boldsymbol{N}}_{gg} = 2\boldsymbol{I}_4 - 0.5\,\boldsymbol{J}_4$ (1.5 on the diagonal, $-0.5$ off it). The right-hand sides are $\boldsymbol{u} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{y}$: $\boldsymbol{u}_a = (14.5, 6.5)$ (each receiver's pseudoranges summed) and $\boldsymbol{u}_g = -(4, 10, 0, 7)$; then $\bar{\boldsymbol{u}}_g = \boldsymbol{u}_g + (14.5 + 6.5)/4 = (1.25, -4.75, 5.25, -1.75)$, which sums to zero.
2. $\boldsymbol{J}_4$ has eigenvalue 4 along $\mathbf{1}$ and 0 on its complement, so $\bar{\boldsymbol{N}}_{gg}$ has $2 - 0.5 \times 4 = 0$ along $\mathbf{1}$ and 2 three times. The normalized null eigenvector is $(0.5, 0.5, 0.5, 0.5)$: weight $1/\sqrt{4}$ on every satellite clock.
3. Row $i$ reads $2g^i - 0.5\,S = \bar{u}_i$ with $S = \sum g$. With $g^1 = 0$ the first row gives $S = -2.5$, then $g = (0, -3, 2, -1.5)$. Back-substitution $a_r = (u_{a,r} + S)/4$ gives $a = (3, 1)$, the solution of Question 2.2. *Checked.*

**Why it works.** Elimination projects each station's information onto the shared unknowns without losing any: it is exact. It cannot create information either, so the shared null direction survives, seen now from the satellites alone.

**Pitfall.** Reducing $\boldsymbol{N}$ but not the right-hand side, or solving the reduced system without a reference choice: it is still singular. That $\bar{\boldsymbol{u}}_g$ sums to zero is the consistency condition that makes a solution exist.

**Takeaway.** The real network does this per station every epoch (Equation (5)), and the weight $1/\sqrt{m}$ is what to look for in its rank gate (Questions 3.2 and 3.5).

*Source.* Equation (5) (status report, step 6); the worked example's data.

**Answer 3.2** (question, p. 8)

**Strategy.** Try constant shifts on subsets of the unknowns. Each independent family that can absorb a common shift gives one null direction.

**Solution.**

1. Two. The first is the old clock direction: every receiver clock and every satellite clock, GPS and Galileo, $+\kappa$ (each Galileo row still holds $a_r - g^i$). The second exists because of the inter-system bias: $d_r^E + \kappa_E$ at every receiver and $g^i + \kappa_E$ for every Galileo satellite leaves every row unchanged.
2. Two. After the receiver terms are eliminated they are spanned by a flat shift of the GPS clocks and a flat shift of the Galileo clocks. When both eigenvalues are zero an eigensolver may return any rotation of the pair.
3. $1/\sqrt{32} = 0.1768$ on every GPS clock and $1/\sqrt{24} = 0.2041$ on every Galileo clock. A small numerical check (two receivers, three GPS and two Galileo satellites) has 9 unknowns, rank 7, and a two-dimensional reduced null space constant within each constellation. *Checked.*

**Why it works.** The inter-system bias gives each receiver a separate clock for Galileo, so the Galileo clocks get their own common level.

**Pitfall.** Expecting the two near-zero eigenvectors to come out pure per constellation on every epoch; a degenerate pair may mix.

**Takeaway.** The declared dimension of the gate is the number of constellations: 2 for GPS and Galileo (Questions 3.5 and 4.3).

*Source.* the arithmetic; the real run in Question 3.5.

**Answer 3.3** (question, p. 8)

**Strategy.** Group the six terms of Question 2.4 by $(n-1)$ and by $m$; then substitute.

**Solution.**

1. $1 + 2f + (n-1) + m + f(n-1) + f\,m = 1 + 2f + (1+f)(n-1+m)$; the slant ionosphere adds $n - 1 + m$.
2. With $f = 2$ and $m = 32$: $5 + 3(n+31) + (n+31) = 5 + 4(n+31)$, which gives 369 for $n = 60$, 361 for 58 and 1,349 for 305.
3. $1 + 4 + 3 \times 115 = 350$, plus 115: 465. *Checked.*

**Why it works.** Each added station brings $1 + f$ directions (its clock against its biases, and one per frequency between its phase bias and ambiguities), plus one for its slant ionosphere: four at $f = 2$, the slope of the figure.

**Pitfall.** Using the single-system formula as an exact count for two constellations; the record has no two-system form, so 465 sets the scale only.

**Takeaway.** Eq. 24 counts every direction; the gate counts what is left after the lumps, pins and priors (Question 3.5).

*Source.* `SBASIS-THEORY-E.md` lines 106 to 114; deck slide 6.

*Check.*

```
C:/Python313/python.exe - <<'EOF'
f = 2
for n, m in ((60, 32), (58, 32), (305, 32), (60, 56)):
    print(n, m, 1 + 2*f + (1 + f)*(n - 1 + m) + (n - 1 + m))
EOF
```

**Answer 3.4** (question, p. 9)

**Strategy.** Take the terms of Equations (1) and (2) one by one and find the row that holds each; then look for a term with no row.

**Solution.**

1. Shared rows: satellite clock (GPS, Galileo), satellite phase bias (GPS L1C, L2W; Galileo L1C, L5Q), satellite code bias (GPS C1C, C2W; Galileo C1C, C5Q). Local rows: ambiguity per signal, slant ionosphere per system, zenith wet delay, receiver clock, Galileo inter-system bias, receiver DCB per system; receiver position "pinned: held at the roster coordinate". The map: $\mathrm{d}t_r$ to receiver clock, $\delta t_r$ to inter-system bias, $T$ to zenith wet delay, $I$ to slant ionosphere, $B_{r,s}$ to receiver DCB, $N$ to ambiguity, the satellite terms to the shared rows. The receiver phase bias has no row (`rx_phase_bias no no`, line 380): the ambiguities absorb it, as in Question 1.2.
2. Shared $(1 + 2f)\, m = 5 \times 56 = 280$ (line 639, `nshared=280`). Local: 3 per line of sight (two ambiguities, one slant ionosphere) for 891 lines of sight, plus 5 per station for 60 stations: $2{,}673 + 300 = 2{,}973$. *Checked.*

**Why it works.** The table is the network's own statement of the model: every parameter it estimates, and where it lives. The missing receiver phase bias is a lump of Question 2.4 made by design.

**Pitfall.** Reading the border counts (498, 393) as unique columns: they count one instance per line of sight. The unique shared columns are 280.

**Takeaway.** Every term of the user's equations has a home in the network, shared or local; the arrowhead of Figure 2 at the scale of 60 stations.

*Source.* `runs/_rf0hv15_sim6/stdout.txt` lines 380, 639, 658 to 686; `src/core/network/assembly/ParamInventory.cpp` lines 645 to 661.

**Answer 3.5** (question, p. 9)

**Strategy.** Find the sub-cutoff eigenvalues and their eigenvector weights, compare the weights with Question 3.2, then express the gaps around the cutoff in powers of ten.

**Solution.**

1. At tow 259230: $+4.036 \times 10^{-11}$ and $-1.803 \times 10^{-10}$, two zeros up to rounding (one is negative). Their eigenvectors put 0.2040 on every Galileo clock and 0.1767 on every GPS clock: the flat shifts predicted by Question 3.2.
2. Cutoff $= 10^{-11} \times 1.0459 \times 10^{5} = 1.0459 \times 10^{-6}$. Cutoff over the larger zero: $1.0459 \times 10^{-6} / 1.803 \times 10^{-10} = 5.8 \times 10^{3}$, 3.8 decades. Weakest real direction (a GPS L2W phase-bias direction) over the cutoff: $1.124 \times 10^{-4} / 1.0459 \times 10^{-6} = 107$, 2.0 decades. The rule, "cutoff $\geq$ 2 decades above the gauge depth and $\geq$ 2 decades below the weakest-real layer", holds, the upper margin only just. The run end (line 4899) reports 720 epochs at tol $10^{-11}$, all emitted, observed count min 2, max 2.

3. Lumps and rows removed them before the gate, as in Question 2.4. The record maps types to rows: 1a to the 2 clock pins (which act after the gate); 3a and 3b to 112 code-bias rows; 5 to 112 ambiguity pins; 1b, 2a, 2b and 4 (240) to the local receiver side (free clock, inter-system bias, DCB with 120 prior entries, no receiver phase-bias state). $2 + 112 + 112 + 240 = 466 = 465 + 1$, the extra one being the second constellation's clock direction. *Checked.*

**Why it works.** The gate separates two populations of eigenvalues, true zeros and weak but real directions, by a threshold placed in the gap between them. The gap must be wide on both sides, or rounding moves an eigenvalue across it.

**Pitfall.** Treating the tolerance as neutral. At the loader default $10^{-9}$ the cutoff here would be $1.05 \times 10^{-4}$, within 7 % of the weakest real direction; the ledger records that $10^{-9}$ withholds 4 real-data epochs.

**Takeaway.** The rank gate confirms one clock direction per constellation every epoch; what it does not see is the subject of Question 4.5.

*Source.* `rf_null_space_read.py` output; `runs/_rf0hv15_sim6/logs/network_estimability.csv` rows 2 to 6; `runs/_rf0hv15_sim6/stdout.txt` lines 640, 4899; `RankGate.cpp` lines 408 to 410; `SBASIS-CONTRACT.md` lines 59 to 68; `docs/CONFIG_promotion_ledger.json` (row `rank_gate_tol`).

## Lesson 4: the project's reference choices

**Answer 4.1** (question, p. 10)

**Strategy.** Find the box, then translate it into the six-unknown example of Question 2.2: what was the same in both solutions, and why?

**Solution.**

1. An estimable function $\boldsymbol{f}^{\mathrm{T}}\boldsymbol{x}$ ($\boldsymbol{f}$ in the range of $\boldsymbol{A}^{\mathrm{T}}$) "is invariant in value AND variance under any admissible datum": under every reference choice that removes exactly the null space. Question 2.2 showed the value half, four differences equal in two choices.
2. Position, zenith delay and vertical ionosphere are estimable before any choice (their rows of the null-space basis are zero). A satellite clock and a satellite phase bias are not: "The satellite clock does not exist as an estimable number; only a datum-dependent lump does."
3. The differential wander was itself estimable, so no reference choice could move it; every datum comparison came out negative. "The cure must alter ESTIMATION (weights / dynamics / model / transmitted covariance), not the gauge." The coverage change that cut the wander from 47 to 14 cm was such a change.

**Why it works.** $\boldsymbol{f}$ in the range of $\boldsymbol{A}^{\mathrm{T}}$ means $\boldsymbol{f}^{\mathrm{T}}\boldsymbol{V} = \boldsymbol{0}$ for the null space $\boldsymbol{V}$. Two admissible solutions differ by $\boldsymbol{V}\boldsymbol{\beta}$, so $\boldsymbol{f}^{\mathrm{T}}\boldsymbol{x}$ is the same for both; the variance argument runs the same way through the covariance.

**Pitfall.** Hunting for a better datum to cure a problem in an estimable quantity. A datum can only relabel; a real problem needs a real change of estimation.

**Takeaway.** A reference choice decides how results are reported, never what the data say: the basis for grading only estimable quantities (Section 5).

*Source.* `SBASIS-THEORY-E.md` lines 40 to 41, 75 to 83, 116 to 120, 223 to 225.

*Check.*

```
sed -n '75,83p;116,120p' docs/cards/network-vehicle/SBASIS-THEORY-E.md
```

**Answer 4.2** (question, p. 11)

**Strategy.** Run the verify, then grep the configuration for each choice's keys and match each to its two-receiver analogue.

**Solution.**

The verify prints, per reference configuration, "76 recommended knobs armed in regime, 2 unarmed (2 documented exceptions), 0 failure(s)" and "golden verify: PASS". *Checked.*

1. Clock pin: no key (the default), with `sticky_pivot` (line 324) and `handover_stransform` (343). Anchors: `phase_datum_prioritized` (297), `phase_datum_held_basis` (925), `ambiguity_vehicle float_kf` (924). Code-bias datum: `estimate_sat_code_bias` (293) and `sat_gf_datum broadcast_tgd` (922); the ionosphere-free pin and the sum-to-zero rows have no key of their own. Receiver DCB prior: `rx_code_bias_datum` (292).
2. The pin (analogue $g^1 \equiv 0$) holds until the pivot leaves, then the constellation is re-referenced: one handover per constellation in 6 h (E02 to E03, G01 to G02). The anchors (analogue $N_A^i \equiv 0$) are a single point of failure: a handover steps the bias by whole cycles unless the successor is pinned at its own level (308 basis values elected, 22 of them cold, declared). The sum-to-zero rows would move at every rise and set, as in Question 2.6; the simulated case suppresses them because the pins already fix every channel (2,880 suppressed $= 4 \times 720$). The ensemble was "SUPERSEDED BY MEASUREMENT" and deleted (commit 7914263).
3. `clock_datum_sum_to_zero`, `clock_datum_broadcast_anchored` and `pivot_election` are absent; `phase_datum_eq34` is false (line 345).

**Why it works.** Each choice removes one family of null directions from Section 3: the clock pin the clock directions, the anchors the per-satellite phase directions, the code-bias datum the satellite code-bias directions, the DCB prior the receivers' geometry-free direction.

**Pitfall.** Concluding from a missing key that a choice is absent: the clock pin is the default and has none.

**Takeaway.** Four choices, each a lesson 2 move at scale; the next question counts their rows.

*Source.* `SBASIS-THEORY-E.md` lines 143 to 150, 239 to 259; `Sbasis.hpp` lines 162 to 166; `config/golden/golden_sim_refound.json`; `_rap_scb_g_q6/stdout.txt` lines 4385, 4400, 4845.

*Check.*

```
C:/Python313/python.exe scripts/make_golden_config.py --verify
grep -n -e '"sticky_pivot"' -e '"phase_datum_held_basis"' -e '"phase_datum_eq34"' \
    config/golden/golden_sim_refound.json
```

**Answer 4.3** (question, p. 11)

**Strategy.** Count the shared columns per satellite, then ask which columns each choice fixes and whether it acts before or after the gate.

**Solution.**

1. $56\times(1+2+2) = 280$ shared columns. Clock pins 2 (E02, G01). Ambiguity pins $112 = 32\times2+24\times2$, one per phase-bias column, placed on 18 stations. Code-bias rows 112: 56 ionosphere-free pins and 56 geometry-free pins at $T_{GD}$ (54 at $cT_{GD}$, 2 with a broadcast $T_{GD}$ of 0); sum-to-zero 0. Receiver prior entries $120 = 60 \times 2$, each of information $1/(10\text{ m})^2 = 0.01\text{ m}^{-2}$. Total: 226 rows on the shared state plus 120 prior entries.

2. The ambiguity pins and code-bias rows act before the gate and fix the 224 phase-bias and code-bias columns. The 56 clocks keep one flat direction per constellation (Question 3.2); the clock pins act in the solve after the gate, so the gate expects exactly 2 ("exactly #constellations residual deficiencies").
3. On a mismatch the gate throws; the heal soft-anchors near-null phase-bias columns ($\sigma$ 1 m) and re-gates once; if the count still differs the epoch is withheld and counted as under- or over-specified. Completion would first fold one eigenvector row per extra separable near-null direction; it folded none in 720 epochs. *Checked.*

**Why it works.** Placing the clock pins after the gate turns the gate into a test: if anything other than the two clock directions is free, the count is wrong.

**Pitfall.** Reading the gate as a proof that the reference choices are correct. It counts null directions; a row that pins something estimable removes a direction that was never null, and the count still matches.

**Takeaway.** The choices are 226 rows and 120 prior entries, not "a few"; the gate checks their number every epoch, not their harmlessness.

*Source.* `_rap_scb_g_q6/stdout.txt` lines 631, 679 to 738, 4109, 4112, 4387; `Sbasis.hpp` lines 50 to 55; `RankGate.cpp` lines 193 to 305; `SBASIS-CONTRACT.md` lines 87 to 88.

*Check.*

```
sed -n '631p;679,738p;4109,4112p' \
    D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/stdout.txt
```

**Answer 4.4** (question, p. 11)

**Strategy.** Use the split of Question 1.1 at the satellite: a pair whose ratio is $\mu_2$ has a zero ionosphere-free part and only a geometry-free part. Compare that part with the broadcast group delay.

**Solution.**

1. `B,G05,C1C,-3.2108400641001413` and `B,G05,C2W,-5.288075205569585`: the ratio is $1.646944 = \mu_2$, so the ionosphere-free combination is zero. The navigation record gives $T_{GD} = -1.071020960808 \times 10^{-8}$ s, and $cT_{GD} = -3.2108400641$ m, the C1C record. The pinned pair carries no estimate: its ionosphere-free part is pinned to zero and its geometry-free part to the broadcast group delay (54 satellites at $cT_{GD}$, 2 with a broadcast value of 0; stdout line 631).
2. With a slant ionosphere the satellite's true geometry-free code bias is not estimable: it merges into the slant ionosphere of every line of sight to the satellite (Question 1.3, mirrored at the satellite). The corrections carry no ionosphere record, so a user estimates its own slant ionosphere, which absorbs the same lump: code biases and ionosphere act as a pair. *Checked.*

**Why it works.** The two code biases of the pinned pair are exactly the two directions of Questions 1.1 and 1.3 at a satellite, and the datum fixes both: the ionosphere-free one to the clock convention, the geometry-free one to $T_{GD}$.

**Pitfall.** Interpreting the published B records of the pinned pair as estimated hardware biases, or comparing them with another provider's DCBs as a check of the network.

**Takeaway.** The satellite clock is therefore the ionosphere-free clock of Question 1.1, and the network's slant ionosphere carries the satellites' true geometry-free biases.

*Source.* `_rap_scb_g_q6/products.txt` lines 61 to 62; `data/2024-129/products/BRDC00IGS_R_20241290000_01D_MN.rnx` line 568; `docs/refound/RF-0c_gf_pin_2026-09-15.md` lines 22 to 23; `SBASIS-CONTRACT.md` lines 114 to 133.

*Check.*

```
C:/Python313/python.exe - <<'EOF'
print(299792458 * -1.071020960808e-08)          # c * T_GD of G05, m
print(-5.288075205569585 / -3.2108400641001413)  # C2W / C1C
print((1575.42 / 1227.60) ** 2)                 # mu_2
EOF
```

**Answer 4.5** (question, p. 12)

**Strategy.** Information along a unit direction is the sum, over the rows that touch it, of (loading$^2$)/$\sigma^2$; the standard deviation is one over its square root.

**Solution.**

1. $0.0074/10^2 = 7.4 \times 10^{-5}$ m$^{-2}$; $\sigma = 1/\sqrt{7.4 \times 10^{-5}} = 116$ m.
2. $(0.0074 + 0.2392^2 + 0.1334^2)/0.0074 = (0.0074 + 0.0572 + 0.0178)/0.0074 = 11.1$, about eleven times, $\sigma$ about 35 m: still no constraint worth the name. Card R1D found such a row changes nothing at 10 m, and at 1 m shrinks the move only by a fifth while pushing 0.95 m into C1W rows: "the datum row is not the fix".
3. $\sigma = 1/\sqrt{110} = 0.095$ m. The carried ambiguity coupling supplies 109 of the 110 and the DCB prior $7 \times 10^{-5}$: the coupling to the shared state holds the level.
4. The direction lives inside one station's block, which is eliminated before the shared matrix is formed (Question 3.1), so the gate never sees it. MAJU00MHL anchored phase biases of five GPS satellites (13 pins). With the pins held, the station's level plus a matching shift of its anchored phase biases is invisible to its data (residual $2.8 \times 10^{-11}$), so a move of the level reaches the published phase biases. In the GPS L5 event the level moved $-21.92$ m; barring the station from the anchor election removed the event: "The anchor loop is the transmission". The hold card ALD derives is not built; GPS L5 stays sequenced.

**Why it works.** A weak direction is safe only while nothing pushes along it. An anchor station couples its local level to the shared phase biases, so a push on the level reaches every user.

**Pitfall.** Adding datum rows on components with small loadings: the information they add scales with the loading squared.

**Takeaway.** The gate checks the shared null space; local weak directions need their own instrument. Lesson 5 turns to what a user receives.

*Source.* card R1D lines 21 to 28, 54, 66 to 89, 99 to 122, 159; card ALD lines 27, 31 to 41, 54, 58 to 65.

*Check.*

```
C:/Python313/python.exe - <<'EOF'
import math
print(1 / math.sqrt(0.0074 / 100))                  # sigma from the prior, m
print((0.0074 + 0.2392**2 + 0.1334**2) / 0.0074)     # information factor
print(1 / math.sqrt(110))                           # sigma from the coupling, m
EOF
```

## Lesson 5: what a user sees

**Answer 5.1** (question, p. 12)

**Strategy.** For each change, find the user unknown whose coefficient pattern matches it: common to all satellites (receiver clock), constant in one satellite's phase rows (that ambiguity), or proportional to $\mu_s$ on code (that slant ionosphere). The answer of Question 2.5 is the first two cases in numbers.

**Solution.**

1. $\Delta$ enters every code and phase row of system $g$ identically, so the receiver clock (with the inter-system bias for a second system) takes it up, as $\hat{a}_C$ took up $a_A$. The geometry terms, hence the position, see nothing.
2. A constant $d$ enters only that satellite's phase rows, as a constant: the float ambiguity moves, $N \to N + d/\lambda$. The position is unchanged; only the ambiguity's meaning changes, which matters once a user fixes integers.
3. $(\delta, \mu_2\delta)$ adds $\mu_s\delta$ to code row $s$, the ionosphere's own signature, so $I^i \to I^i + \delta$ absorbs it; on phase that shift enters as $-\mu_s\delta$, which the float ambiguities absorb. The position is unchanged, the user's slant ionosphere biased by $\delta$. Its ionosphere-free combination is zero, so an ionosphere-free user sees nothing at all.
4. The per-system mean at each epoch (case 1, the receiver clock), then each satellite's mean over the session (case 2, the ambiguity). Case 3 does not arise for clocks.

**Why it works.** These are the three null directions of Section 1, now at the user's receiver. A change of the corrections along them is a change of the user's own unknowns; the S-system version: the rank-one "$[e; e]$ (clock/geometry) and $[-\mu; \mu]$ (iono) parts lie in the user's own design range and are absorbed".

**Pitfall.** Comparing two sets of corrections satellite by satellite without removing these parts; the differences can be large and mean nothing for a user.

**Takeaway.** Grade only datum-invariant or single-difference quantities (standing invariant 5); the two means of the clock comparison are that rule applied.

*Source.* `SBASIS-THEORY-E.md` lines 158 to 182; `SBASIS-CONTRACT.md` lines 98 to 102, 114 to 133; status report, step 5; `results/t3/results.md` lines 64 to 66.

**Answer 5.2** (question, p. 13)

**Strategy.** Open the writer and read off, in order, what each comma-separated field is; then sort the fields into reference choices and other declarations.

**Solution.**

| Field | Declares | Defined or written at |
|---|---|---|
| `S30000SIM` | the run's reference (clock-datum) station; the pins act on satellite clocks | `ApplicationNetworkSetup.cpp` line 1485 |
| `1`, `nominal-wu93` | bias sign convention; phase wind-up model | `Sbasis.hpp` lines 98 to 108, 215 |
| `0` (5th field) | ambiguity datum: "legacy single pivot and prioritized anchor" | `Sbasis.hpp` lines 129 to 132 |
| `0` (6th field) | clock datum: one hard satellite clock pin per constellation | `Sbasis.hpp` lines 162 to 166 |
| `1`, `0` | the network emits its own code biases; code-bias datum id | `Sbasis.hpp` lines 229, 238 |
| `0.05`, `1` | clock discontinuity threshold (m); discontinuities emitted | `NetworkProductEmitter.hpp` lines 195 to 207 |
| the SP3 and CLK names | the a priori orbit and clock; every C record corrects this clock | `ApplicationNetworkSetup.cpp` lines 1524 to 1566 |
| `E:E02`, `G:G01` | realized clock pivots (C = 0 at the first epoch) | `SolPrep.cpp` lines 1176 to 1179 |
| `cbjump`, `atx`, `oref` | code-bias jump test; antenna file; orbits at the centre of mass | `NetworkProductEmitter.hpp` lines 222 to 228; `NetworkProductIO.hpp` lines 270 to 273 |
| `discfrac`, `hsig` | phase jump gate of 0.25 cycle; $\sigma_0$ of the H records | `NetworkProductEmitter.hpp` lines 238 to 248; `NetworkProductIO.hpp` lines 238 to 247 |
| `cbpin=...` | the ionosphere-free pin pair per system | `SolPrep.cpp` lines 1118 to 1157 |
| `sat_gf_datum=broadcast_tgd` | the geometry-free datum at $cT_{GD}$ | `SolPrep.cpp` lines 1165 to 1172 |
| `pbhold=2` | the phase-clock hold (Question 6.3) | `ApplicationNetworkSetup.cpp` lines 1599 to 1601 |

The reference choices of Section 4 are the two datum ids, the pivots, the `cbpin` pairs and `sat_gf_datum`; `pbhold` declares how the phase biases are held. A current run appends `wumode=accumulated`. The real-data S line declares only `cbpin=G:C1C-C2W`: the Galileo pairs are "NOT declared (per-satellite split)" (`runs/_rf0hv15_field6/stdout.txt` line 686).

**Why it works.** A user can absorb a reference choice without knowing it (Question 5.1), but a grader, another network or a user that fixes integers must know which choice was made. The header line is the network's declaration of the conventions behind every number in the file.

**Pitfall.** Assuming every choice is declared because the line exists: on real data the Galileo pin pairs and the anchor stations are not named.

**Takeaway.** The declaration is how lesson 2's "the producer declares its datum" looks in a real file.

*Source.* `src/core/products/NetworkProductIO.cpp` lines 465 to 602 and the lines in the table (paths under `src/core/network/` and `include/`).

*Check.*

```
sed -n '2p' D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/products.txt
```

### Answer 5.3 (question, p. 13)

**Strategy.** Separate the step into the part common to all GPS satellites (the mean) and the part left on each satellite (the spread about it); then apply Question 5.1 to each part.

**Solution.**

1. The awk prints line 134213, `E,2313,271230.00000715256,400,0,0,1,E` (Galileo, 03:20:30), and line 189631, `E,2313,276149.99999547005,564,0,0,1,G` (GPS, 04:42:30). The script prints "31 satellites, mean step 1.6278 m, sd after the mean 0.0027 m". For comparison, an ordinary pair of epochs (276090 to 276120) gives −0.0001 and 0.0008 m.
2. The handover of Question 2.6, part 3: the pivot moved from G01 to G02 and every GPS clock stepped by the old pivot-relative value. A user's receiver clock absorbs the common 1.63 m; 3 mm per satellite stays visible, a few times the ordinary epoch-to-epoch spread.
3. A raw-step grade sees 31 satellites jump 1.6 m at once; the datum-invariant step is 3 mm. On real data the GPS group mean steps by about +154 cm at its pivot change. *Checked.*

**Why it works.** A pin moves every clock of its constellation by the same amount when its satellite leaves; the common amount is a reference change, the per-satellite remainder is what the data actually say.

**Pitfall.** Counting clock steps per satellite in raw corrections and calling a declared handover a jump defect.

**Takeaway.** Remove the per-system mean before grading clocks; the toy's +3 m handover and this 1.63 m step are the same event.

*Source.* `runs/_rf0hv15_sim6/products.txt`; `results/t3/results.md` lines 105 to 107.

**Answer 5.4** (question, p. 14)

**Strategy.** Write each chord as a difference of two independent errors and use $\mathrm{Var}(x-y) = \mathrm{Var}(x) + \mathrm{Var}(y)$ and, for two chords sharing $y$, $\mathrm{Cov} = \mathrm{Var}(y)$.

**Solution.**

1. A chord is $e_i = s_i - s_{\mathrm{ref}}$: $\mathrm{Var}(e_i) = \sigma^2 + \sigma^2_{\mathrm{ref}}$ and $\mathrm{Cov}(e_i, e_k) = \sigma^2_{\mathrm{ref}}$, so two chords correlate by $\sigma^2_{\mathrm{ref}}/(\sigma^2 + \sigma^2_{\mathrm{ref}})$: 0.5 when $\sigma_{\mathrm{ref}} = \sigma$, and more when the reference satellite is the worst one.
2. L1C over all epochs: as traced, median $|e|$ 0.304 cycle and 65.3 % round to the right integer; reference-free 0.122 cycle and 81.6 %. Last quarter: 0.260 cycle and 69.5 % against 0.092 cycle and 86.5 %.
3. The reference satellite's error and its returns enter every chord, so a grade on the traced chords reports that one element's share: "Quote DD grades ref-free from now on." *Checked.*

**Why it works.** Removing the median over a group of chords that share a reference removes the reference's error, which is common to the group, and keeps each satellite's own.

**Pitfall.** A related trap of the same card: subtracting a group's own circular median shrinks pure noise from 0.251 to 0.145 cycle, so a wrapped grade is quoted as a gap to a size-matched null, never as a raw number.

**Takeaway.** Double differences are estimable, but a grade built on one reference element inherits that element; grade reference-free.

*Source.* `docs/cards/sim-bed/DD-TRACE-REF-SAT.md` lines 41 to 48, 107 to 110, 143 to 159; `CLAUDE.md`, "Grading traps that still apply".

## Lesson 6: the satellite clock

**Answer 6.1** (question, p. 14)

**Strategy.** A random walk with spectral density $q$ has variance $q\,t$ after time $t$; a drift of rate $r$ has moved $r\,t$. Compare the two as functions of time.

**Solution.**

1. $\sqrt{2.9 \times 10^{-8} \times 30} = 0.933$ mm per epoch; $\sqrt{2.9 \times 10^{-8} \times 21{,}600} = 2.50$ cm over 6 h. The value is measured: "$q_{\text{phys}} \approx 2.9 \times 10^{-8}$ m$^2$/s $\rightarrow$ 0.94 mm per 30 s".
2. After $n$ epochs the drift is $0.30n$ mm and the walk's $1\sigma$ $0.933\sqrt{n}$ mm; they are equal at $n = (0.933/0.30)^2 = 9.7$ epochs, about 5 min. At 6 h the drift is 21.6 cm, 8.6 times the walk.

**Why it works.** A step standard deviation is being compared with a signed, persistent increment. The drift grows as $n$, the walk's spread as $\sqrt{n}$, so a walk of this density cannot follow a sustained rate however small the rate per epoch looks. Card SCB reads the outcome as partial tracking.

**Pitfall.** Judging a process-noise value by the size of one epoch's step against one epoch's signal.

**Takeaway.** $q$ sets how fast the estimate may follow the truth; the next question turns that into a time constant.

*Source.* `config/golden/golden_sim_refound.json` line 323; `include/network/solve/ClockAnchor.hpp` lines 35 to 36; `docs/cards/clock/Q3-PHYSICAL-FLOOR.md` lines 29 to 32; card SCB lines 29 to 35.

**Answer 6.2** (question, p. 15)

**Strategy.** A steady-state filter averages about $1/K$ epochs: the time constant is $\Delta t/K$, and a ramp leaves an error of rate times time constant.

**Solution.**

1. $R = 0.0923^2/2 = 4.26 \times 10^{-3}$ m$^2$; $K = \sqrt{8.7 \times 10^{-7}/4.26 \times 10^{-3}} = 0.0143$; $\Delta t/K \approx 2{,}100$ s, about 35 min (the exact steady state of the scalar filter gives 2,084 s).
2. $10^{-5} \times 2{,}100 \approx 2.1$ cm. Card SCB measured 2,127 s (2,000 s GPS only): the anchor's $q$ sets the time constant, within about 2 %.
3. $K = \sqrt{1.5 \times 10^{-4}/4.26 \times 10^{-3}} = 0.19$, about 160 s (exact steady state 146 s), the scale of the measured 52 s: the regime's larger $q$ follows the drift within the epoch noise. *Checked (arithmetic).*

**Why it works.** The gain balances trust in the model against trust in the data: a small $q$ against noisy epochs gives a small gain, a long memory and a long lag.

**Pitfall.** Two cautions. The argument is scalar and stationary, and the real filter is neither. And the grader's error is estimate minus truth with a positive coefficient, which a pure lag would make negative; the card calls it a lag without explaining the sign, so the safe wording is "an error proportional to the drift rate, with a coefficient of about 35 min".

**Takeaway.** A precise clock needs either a precise a priori clock to follow, or a larger $q$ that tracks the truth; the self-contained regime takes the second route.

*Source.* `docs/cards/clock/Q3-ANCHOR-DESIGN.md` lines 47 to 56; card SCB lines 124, 128; `docs/cards/raim-qc/DAY129-FRAME-ANATOMY.md` lines 139 to 143; `scripts/simnet_clock_truth_grade.py` line 151.

*Check.*

```
C:/Python313/python.exe - <<'EOF'
import math
R = 0.0923**2 / 2
for q in (2.9e-8, 5e-6):
    K = math.sqrt(q * 30 / R)
    print(q, 'K %.4f' % K, 'tau %.0f s' % (30 / K), 'error %.4f m' % (1e-5 * 30 / K))
EOF
```

**Answer 6.3** (question, p. 15)

**Strategy.** Substitute the held $P(t)$ into $C - P$ and simplify; then write what the user's phase row contains with the a priori clock $a(t)$ underneath.

**Solution.**

1. $C - P = c(t) - p(t_0) - [\tilde{c}(t) - \tilde{c}(t_0)] = (c - p)(t_0) + M(t) - M(t_0)$: constant per satellite up to the system mean $M$, which the receiver clock absorbs. On code the user applies $a(t) + C(t)$, the network's estimate; on phase it sees $a(t) + C(t) - P(t)$, which is $a(t)$ plus a constant: the a priori clock's motion. With truth $a(t) + e(t)$, the phase rows are wrong by $-[e(t) - e(t_0)]$ plus a constant; the float ambiguity absorbs the constant, the change passes one to one.
2. Mean square over hour 6 ($t$ from 18,000 to 21,600 s): rate part $\sigma_r^2(t_{\mathrm{hi}}^3 - t_{\mathrm{lo}}^3)/(3 \times 3600) = 10^{-10} \times 3.931 \times 10^8 = 0.0393$ m$^2$; walk part $q(t_{\mathrm{lo}} + t_{\mathrm{hi}})/2 = 0.0006$ m$^2$; rms 0.200 m, against the measured 0.201 m (GPS) and 0.206 m (Galileo).
3. The loader refuses `network.emit_pb_hold_phase_clock` with `network.apriori_source sp3_brdc_clock`: "the phase-clock hold is valid only with a precise a priori clock product"; with `brdc_rts` it only warns. Part 1 is the reason: the hold hands the user the a priori clock's motion. The S line declares `pbhold=2` (3 for the smoothed hold of the self-contained postures). *Checked.*

**Why it works.** Holding $C - P$ makes the phase-visible clock follow the a priori clock and cancels the network's own clock motion on the phase rows, where the network's correction is noisy (Question 9.3). The price is that the a priori clock's errors pass through untouched.

**Pitfall.** Reading "the phase bias is held constant" literally: the phase-bias record moves with the estimated clock; it is the difference $C - P$ that is held. On real data, ignoring the precondition cost the user 1.191 and 0.743 m in hours 1 and 2, against 0.108 and 0.024 m with CODE.

**Takeaway.** A user's phase precision is the a priori clock's: this is why the network's clock precision is CODE's today.

*Source.* `include/network/emit/PbHold.hpp` lines 60 to 66; card LAH lines 30 to 36; card SCB lines 129 to 145; `src/core/config/GnssConfigLoadNetwork.cpp` lines 311 to 319, 2612 to 2617; card SCL lines 22 to 26, 76 to 78.

*Check.*

```
C:/Python313/python.exe - <<'EOF'
lo, hi = 18000.0, 21600.0                       # hour 6, s after the reset
rate = 1e-10 * (hi**3 - lo**3) / (3 * 3600)     # rate part, m^2
walk = 3e-8 * (lo + hi) / 2                     # random-walk part, m^2
print((rate + walk) ** 0.5)
EOF
sed -n '311,319p' src/core/config/GnssConfigLoadNetwork.cpp
```

**Answer 6.4** (question, p. 16)

**Strategy.** Read the two outputs side by side, with Question 6.2 (which posture should follow a drift) and Question 6.3 (what reaches the user) as predictions to check.

**Solution.**

Reference case `_rap_scb_g_k6`: clock level at hour 6 0.0533 m; per-satellite mean over hours 2 to 6, rms 0.0426 m over 55 satellites; correlation with the drawn rate +0.524; user by hour 0.1976, 0.2012, 0.3189, 0.2142, 0.2370, 0.3376 m. Self-contained regime `_rap_scb_c_k6`: 0.0232 m; 0.0205 m over 31 satellites; +0.026; user 0.2287, 0.1296, 0.1003, 0.0211, 0.0311, 0.0260 m. The regime follows the drift (no correlation with the rate); the reference case does not, as Question 6.2 predicts. The reference case's user error is about ten times its clock error because the hold passes the a priori clock's error to the phase rows whatever the estimator tracks, as Question 6.3 predicts. *Checked.*

**Why it works.** The estimate and the corrections are different things: a good clock estimate helps a user only through the rows the user actually receives it on.

**Pitfall.** Reading more into one realization than it holds: the error statistics were chosen, the regime case is GPS only (31 satellites against 55), the sign question of Question 6.2 is open, and neither the regime nor the native posture is armed. Card SCB's text says the regime and native users "do not move" between data sets; its tables show the native user up to 2.1 cm higher on the clock data set in hours 1 to 3.

**Takeaway.** Clock independence is a model change (the regime and its hold), and it is measured on simulated error only.

*Source.* the two grader outputs; card SCB lines 8, 10, 122 to 145.

**Answer 6.5** (question, p. 16)

**Strategy.** Ask what a median does to a sum, then check the comparison's conditions: which configuration, which days, which hours.

**Solution.**

1. $0.0024 - 0.0007 + 0.0099 = 0.0116$ m, not 0.0171 m: the steps add exactly for each station-day, but the median of a sum is not the sum of the medians. The clock's information is the broadcast-clock step, about 1.0 cm.
2. The native posture against the GPS-only reference configuration of 2026-10-07 (GPS and Galileo since 2026-10-13), days 130 to 132 (9 to 11 May 2024), 16 stations, as the median paired change of each station's hours 2 to 6 error; 19 of 48 station-days (39.6 %) are worse by more than 2 cm; day 129 is its most favourable day (+0.3 cm); the first hour adds +7.65 cm.

**Why it works.** Medians describe the typical station-day; they are robust but not additive, so an attribution is read per step, never by adding medians.

**Pitfall.** Quoting "1.7 cm" without its conditions, or as the clock's share: the clock is about 1.0 cm of it, and the comparison predates Galileo.

**Takeaway.** Leaving CODE's clock costs about a centimetre at the median today; that gap is the next step of the program.

*Source.* card NBP lines 170 to 177, 184 to 190, 210 to 219; card GAD.

## Lesson 7: ambiguities

**Answer 7.1** (question, p. 17)

**Strategy.** Treat the two phase rows as a 2-vector in metres. A common cycle on both bands moves it along $(\lambda_1, \lambda_2)$, the ionosphere along $(1, \mu_2)$; the angle between the two directions says how separable they are.

**Solution.**

$\lambda_1 = c/f_1 = 0.190294$ m, $\lambda_2 = 0.244210$ m, $\mu_2 = (77/60)^2 = 1.646944$. $\cos\theta = (\lambda_1 + \mu_2\lambda_2)/(|(1,\mu_2)|\,|(\lambda_1,\lambda_2)|) = 0.592494/(1.926765 \times 0.309597) = 0.99325$, $\theta = 6.66°$. The part orthogonal to the ionosphere is $0.3096 \times \sin 6.66° = 0.0359$ m of the 0.31 m step. "99.3 %" is the cosine, a share of amplitude; as a share of variance ($\cos^2$) it is 98.7 %. *Checked.*

**Why it works.** Two directions 6.66° apart can be told apart only through the small orthogonal part, 3.6 cm per cycle; any ionosphere uncertainty larger than that hides the integer.

**Pitfall.** Treating the percentage as a variance share, and forgetting the metric: it assumes equal phase noise in metres on both bands.

**Takeaway.** With a free slant ionosphere per line of sight, integers are a question of how well the ionosphere is known (Question 7.2).

*Source.* card ARL line 43 (the statement); the arithmetic.

*Check.*

```
C:/Python313/python.exe - <<'EOF'
import math
c = 299792458.0
l1, l2 = c / 1575.42e6, c / 1227.60e6
mu2 = (1575.42 / 1227.60) ** 2
k = (l1 + mu2 * l2) / math.hypot(1, mu2) / math.hypot(l1, l2)
print(k, math.degrees(math.acos(k)), math.hypot(l1, l2) * math.sin(math.acos(k)), k * k)
EOF
```

**Answer 7.2** (question, p. 17)

**Strategy.** An ionosphere error of $d$ metres on band $s$ is $\mu_s d$ metres of phase, which an ambiguity takes as $\mu_s d/\lambda_s$ cycles; a lane takes the difference.

**Solution.**

1. Cycles per metre: L1 $1/\lambda_1 = 5.255$, L2 $\mu_2/\lambda_2 = 6.744$, wide lane $5.255 - 6.744 = -1.489$ (the card: "L1C 5.255, L2W 6.744"). $d = 0.4233$ m: L1 2.22 cycles, wide lane 0.63 cycle. $d = 0.0419$ m: L1 0.22 cycle, wide lane 0.062 cycle.
2. 0.1 L1 cycle = 0.0190 m; 1 TECU $= 40.3 \times 10^{16}/f_1^2 = 0.1624$ m of L1 delay, so 0.1 cycle is 0.117 TECU. The no-pin error, 0.4233 m = 2.61 TECU, is 22 times that: "2.61 TECU, 22× the 0.117 TECU that moves a fraction". *Checked.*

**Why it works.** The wide lane suffers less (1.49 against 5.26 cycles per metre) because the two bands' ionosphere terms partly cancel in it; the narrow L1 integer takes the full effect.

**Pitfall.** Forgetting that 0.4233 m is a per-line-of-sight rms; a double difference of several such terms can be larger.

**Takeaway.** Without integers, the ionosphere is known 22 times too loosely to see L1 integers; a complete set of integers would close the gap (Question 7.4).

*Source.* `docs/cards/raim-qc/T3-IONO-MECHANISM.md` lines 326 to 331; `docs/SPEC_oracle_pin.md` lines 514 to 517, 541 to 546.

**Answer 7.3** (question, p. 17)

**Strategy.** Redo Question 2.5 symbolically, keeping every term; then difference two satellites and see what cancels.

**Solution.**

1. The pins make the network emit $\bar{b}^i = b^i + N_A^i - \zeta^i$ (in cycles), and the user then estimates $\tilde{N}_C^i = N_C^i - N_A^i + \zeta^i = (z_C^i - z_A^i) + (\beta_C - \beta_A) + \zeta^i$.
2. $\tilde{N}_C^i - \tilde{N}_C^k = [(z_C^i - z_C^k) - (z_A^i - z_A^k)] + (\zeta^i - \zeta^k)$: the bracket is an integer double difference (the receiver phase biases cancel), so the difference is an integer exactly when $\zeta^i - \zeta^k$ is. Integer pins (Question 2.5: $\zeta = 0$) give integers; float pins give a fraction.
3. "The WIRE's gauge is an arbitrary receiver's FLOAT ambiguity": the anchors are float targets, re-pinned at the successor's float on every handover, so the user's between-satellite differences carry fractions spread over a whole cycle, and the published biases sit at the uniform null (Question 7.5). Card IRD: a user's single difference is integer if and only if every emitted bias difference is; it proposes to pin a spanning tree of each observation graph ($n_c + m_c - 1$ edges) to integers plus one real-valued root receiver phase-bias pin, so that every chord is an integer (after de Jonge [1]). Not built (IRD-1 is the next step).

**Why it works.** The pins enter the emitted biases and hence every user's ambiguities; only an integer pin passes the integer property on.

**Pitfall.** Concluding that an integer property in the network's own float solution reaches the user: it reaches the user only through the pins.

**Takeaway.** The integer-recovering datum is a reference choice, so it changes no position by itself; its value is that it would make a user's integers fixable.

*Source.* `docs/network_ambiguity_resolution_design.md` lines 335, 352 to 356; card IRD lines 19, 41 to 49.

**Answer 7.4** (question, p. 18)

**Strategy.** Find the row that measures what the data cannot absorb, then compare the no-pin, complete-pin and partial-pin columns of that row only.

**Solution.**

Row 1b, the double-difference-invariant per-line-of-sight L1 slant ionosphere error, carries the deficit: 0.4233 m with no pin, 0.0419 m with every ambiguity pinned (10.1 times smaller; 2.61 to 0.26 TECU). "The `none` column's row 1b IS the integer-information deficit". The raw per-line-of-sight error (2.73 m) is about 97 % station and satellite lumps, which the reference choice and the user absorb. Partial pins do not interpolate: L5Q alone 0.2793 m (better); L1C with L2W 0.5632 m (worse than no pin, although every double difference is then integer); one station 0.4476 m; one satellite 0.4741 m. "Integer information is worth having only when it is complete on the LOS."

**Why it works.** The bands of a line of sight share one ionosphere state, so exact integers on a subset of bands over-determine it; the card calls this a consistent reading, not a proof.

**Pitfall.** Using the raw per-line-of-sight error (Question 5.1: mostly lumps), or comparing these values with today's simulations: they come from a 60-epoch warm resume on `simnet60_quiet_wu93_24h`, a data set whose numbers `CLAUDE.md` marks superseded. The reading stands; the values are not comparable.

**Takeaway.** Integers pay off only when they are complete on a line of sight, which bounds any partial integer scheme.

*Source.* `docs/SPEC_oracle_pin.md` lines 20 to 29, 193 to 195, 514 to 517, 534 to 546, 558 to 568, 611 to 627, 654 to 663.

**Answer 7.5** (question, p. 18)

**Strategy.** Hold the two outputs against Question 7.3: one grades the network's float solution, the other the corrections a user receives.

**Solution.**

First command (4 s): the published GPS phase biases' wide-lane median |fraction| by hour from 00:30:15 is 0.254, 0.260, 0.259, 0.255, 0.253, 0.252, the narrow lane 0.264 to 0.266, against a uniform null of 0.250 (band about 0.22 to 0.27). Second (4 s): "CORRECT-FIX RATE among kept: 5572/5572 = 100.00%", "WRONG FIXES: none". The integers are in the network's float solution (a search on its float states and covariance fixes 5,572 of 13,138 settled double differences of that case, all correct), but the published phase biases carry none of it: they sit at the uniform null, as Question 7.3 predicts for float pins. *Checked.*

**Why it works.** The search reads integers off the network's own states; the corrections carry the float pins on to every user.

**Pitfall.** Attaching the 42 % to the corrections or to the reference configuration: the case is a simpler 2 h simulation without time-correlated phase errors or slips, and with the time-correlated phase error the share is 18.5 %.

**Takeaway.** The integers exist in the float; publishing them needs the integer-recovering datum of Question 7.3.

*Source.* the two outputs; card ARL lines 98 to 102, 142.

## Lesson 8: simulated data

**Answer 8.1** (question, p. 19)

**Strategy.** Ask what the simulator injected and what the network was given as its starting point; a test can only detect an error that is present.

**Solution.**

The quiet data set's truth clock is the CODE final clock, and the network takes the same product as its a priori clock, so the estimable satellite clock truth is identically zero (card SCB, section 1). `results/t2/results.md` confirms it: no `sat_clock` key in the generator's configurations, no ARMED line, no `truth_satclock.csv`. The two `ls` commands print 0 and 61. *Checked.*

**Why it works.** A simulation grades what it injects, plus contrasts in which every reused term cancels; with the a priori clock equal to truth, the clock part of the test is empty.

**Pitfall.** Reading the 0.9 cm agreement as a test of the clock: it "certifies the algebra and the biases, not the clock".

**Takeaway.** The clock-error data set of card SCB exists to fill this gap (Question 6.4).

*Source.* card SCB lines 17 to 25; `results/t2/results.md` lines 123 to 126; `docs/cards/sim-bed/SIMNET-REF.md` lines 269 to 272.

**Answer 8.2** (question, p. 19)

**Strategy.** Convert the offset to seconds and find the first epoch whose elapsed time reaches it; then read the warning log as a count of epochs, keeping in mind which epochs it does not log.

**Solution.**

1. 0.504167 h = 1,815.0012 s; epoch $k$ is $30k$ s after the first, and the first $k$ with $30k \geq 1{,}815.001$ is 61 (1,830 s): tow 261060, 00:31:00. 0.4958333333 h = 1,785 s gives $k = 60$: tow 261030, 00:30:30 (the T2 restart). Each offset sits 15 s, half an epoch, below the epoch it selects, so rounding cannot move the start by an epoch. 00:30:15 (tow 261015) is a grading origin, not a start epoch.
2. Output: 259230 367, 259260 366, 259290 366, 259320 367, 259350 309 (warnings per epoch): 5 epochs logged. A warning needs more than 150 s outside the window, which opens at tow 259500, so 259380 to 259470 are extrapolated without one: 9 epochs, 00:00:30 to 00:04:30. *Checked.*

**Why it works.** The half-epoch back-off keeps the offset away from any epoch boundary, so a tiny rounding difference cannot change which epoch starts the run.

**Pitfall.** Treating "the first 30 minutes" as the extent of the fault: 9 epochs are faulty; what lasts is the state they leave in the network (Question 8.3).

**Takeaway.** Always say whether a number is graded from a run's start or from a grading origin; the two differ (Question 8.3).

*Source.* `include/math/TimeUtils.hpp` lines 416 to 448; `docs/cards/queues/QPR-ITEMS-5-8.md` lines 239, 256; `src/core/model/ModelGeometry.cpp` line 1204; `docs/refound/CTU2_level_recursion_followup_2026-09-29.md` lines 29 to 43.

*Check.*

```
head -2 D:/GnssProcessorSupport/runs_nom/_rap_scb_g_q6/logs/closed_loop_trace.csv
```

**Answer 8.3** (question, p. 19)

**Strategy.** Compare the two runs at the same hours; then ask what the datum share is (Question 4.4) and which side of the comparison it belongs to.

**Solution.**

Baseline (from 00:00:30): level by hour 0.7087, 0.3487, 0.3303, 0.3157, 0.3096, 0.3108 m; per-satellite mean over hours 2 to 6, rms 0.3179 m over 55 satellites (G06 −1.198, E05 +0.872, E36 −0.846). Card SCB's quiet case (from 00:31:00): level 0.0809, 0.0291, 0.0226, 0.0202, 0.0193, 0.0185 m; rms 0.0186 m. With `--no-ds` the baseline's rms reads 0.7784 m. *Checked.*

**Why it works.** The datum share is the ionosphere-free combination of the truth code biases over the pinned pair: the network's clock contains it by construction of the code-bias datum, so it belongs to the truth expressed in the network's basis. Truth is transformed; differencing the estimates instead would bias the variance factor downward, because the estimates share the datum.

**Pitfall.** Grading truth against estimates in different bases, and expecting the user to show the 0.32 m: a float user cannot see a per-satellite constant (Question 5.1).

**Takeaway.** A start inside the extrapolated epochs leaves 0.32 m per-satellite constants in the published clock that no float user notices.

*Source.* the grader outputs; card SCB lines 87, 102, 107 to 109; `scripts/simnet_clock_truth_grade.py` lines 138 to 153; `SIMNET-REF.md` lines 276 to 283.

**Answer 8.4** (question, p. 20)

**Strategy.** Subtract hour by hour (ours minus perfect) to isolate the network's cost; the perfect-corrections column alone is the user's floor.

**Solution.**

Ours minus perfect, in metres: +0.0030, +0.0026, +0.0085, −0.0007, +0.0005, +0.0002. The largest cost is 0.85 cm, in hour 3 (the report's "at most 0.9 cm"); hours 4 to 6 show nothing

measurable. The floor (perfect corrections, hours 4 to 6) is 1.73, 1.81 and 1.76 cm, in the height: median $|U|$ 1.54, 1.54, 1.65 cm against $|E| \leq 0.46$ cm and $|N| \leq 0.55$ cm.

**Why it works.** The same user software with perfect corrections is the control: whatever error remains with them belongs to the user side.

**Pitfall.** Attributing the 1.7 cm vertical floor to the network, or forgetting that hour 6 is 30 min long (the network run ends at 06:00).

**Takeaway.** On the quiet simulation the network's corrections are as good as perfect ones from hour 4; the remaining error is the user's.

*Source.* `results/t2/results.md` lines 195 to 206; `results/t2/metrics.csv`.

## Lesson 9: real data

**Answer 9.1** (question, p. 20)

**Strategy.** Read the two grader outputs as two instruments: the test station by hour, and the median of 16 stations with the paired change; then check the test station's mode and neighbourhood before comparing it with anything.

**Solution.**

`rapid_arm.py score`: `_rf0hv15_field6` median by hour 0.06, 0.01, 0.02, 0.01, 0.02, 0.01 m, 688 epochs below 10 cm; `_rf0hv15_sim6` 0.14, 0.07, 0.03, 0.02, 0.03, 0.02 m, 595. `gcensus.py`: `galp` (GPS and Galileo corrections), 16 stations, median by hour 0.136, 0.054, 0.041, 0.036, 0.040, 0.037 m, WTZZ 0.058, 0.011, 0.018, 0.010, 0.015, 0.011 m; paired against `galpc` (GPS only), hours 2 to 6 median −0.0198 m, 10 better, 2 worse (HARB +0.0158, DAV1 +0.0605), hour 1 −0.2203 m; `galpg` (GPS-only users of the GPS and Galileo corrections) −0.0000 m, ARHT +0.1958 m. Better and worse mean a change of more than 1 cm. The first hour: WTZZ00DEU 5.8 against a mean of 1.3 cm over hours 2 to 6 (4.5 times); the median station 13.6 against 4.2 cm (3.3 times). Mode: `position_q` 100 (moving receiver), `ar_mode` 0 (no integer fixing). None of the 17 user stations is in the 58-station network; WTZZ00DEU shares its site with network stations WTZR00DEU and WTZS00DEU ("the colocated consumer hold-out"; its RINEX header position is about 3 m from WTZR00DEU's). *Checked.*

**Why it works.** One station can be lucky; the median of 16 stations outside the network is the typical user, and the paired change compares each station with itself.

**Pitfall.** Quoting the test station as the typical result, quoting the first hour as a settled result, or treating WTZZ00DEU as unrelated to the network; whether its co-location explains its rank has not been tested.

**Takeaway.** Settled claims come from hours 2 to 6 and from both instruments together.

*Source.* the outputs; `runs/_rf0hv15_field6/arm.json`; `scripts/clock_level/gcensus.py` lines 151 to 153; card GAD lines 70 to 75; `docs/refound/RF-PD_station_rejection_2026-09-15.md` line 87.

*Check.*

```
C:/Python313/python.exe - <<'EOF'
import json
cfg = json.load(open('config/golden/golden_field_refound.json'))
names = [s['name'] if isinstance(s, dict) else s for s in cfg['network']['stations']]
print(len(names), [n for n in names if n.startswith('WTZ')])
EOF
```

**Answer 9.2** (question, p. 21)

**Strategy.** Pool the 25 values (5 days, hours 2 to 6) of each series, compute both a median and a mean, and check day by day where the test station stands.

**Solution.**

The 25 sixteen-station medians have a median of 0.038 m and a mean of 0.0406 m; the 25 WTZZ00DEU values a median of 0.018 m. On day 130 WTZZ00DEU reads 3.9, 5.9 and 5.3 cm in hours 4 to 6 against the median station's 3.5, 3.2 and 3.8 cm. *Checked.*

**Why it works.** A summary over days needs its statistic named: median and mean differ here by 3 mm, and a different pooling gives another value again.

**Pitfall.** Generalizing one day's ranking: "the best station" holds on 8 May 2024 (day 129), not on every day. The report, which first quoted 3.9 cm for the 16 stations, now quotes 3.8 cm (mean 4.1 cm).

**Takeaway.** Multi-day evidence tests whether one day's result is typical; it bounds the wording of the claim.

*Source.* `D:/GnssProcessorSupport/runs_nom/mday_galileo_grades.txt`.

*Check.*

```
C:/Python313/python.exe - <<'EOF'
import re, statistics as st
cen, wtz = [], []
for l in open('D:/GnssProcessorSupport/runs_nom/mday_galileo_grades.txt'):
    m = re.match(r'== mdg\d+c: .*by hour ([\d. ]+); WTZZ ([\d. ]+)', l)
    if m:
        cen += [float(x) for x in m.group(1).split()[1:]]
        wtz += [float(x) for x in m.group(2).split()[1:]]
print(len(cen), st.median(cen), round(st.mean(cen), 4), st.median(wtz))
EOF
```

**Answer 9.3** (question, p. 21)

**Strategy.** List every comparison with its reference and its hours; compare only rows with the same hours, and use Question 5.1 for why two means are removed first.

**Solution.**

Against the GFZ final clock, after the per-system epoch mean and each satellite's mean are removed (STD): our network (CODE final plus c0) GPS 4.3 and Galileo 7.6 cm over 01:00 to 06:00 (6.4 and 11.6 cm including the first hour; Galileo RMS 15.8 and 25.3 cm because of E12); CODE final 1.4 and 2.6 cm over 00:00 to 06:00, 1.2 and 2.2 cm over 01:00 to 06:00, 1.9 and 3.3 cm over 13:24 to 24:00; the IGS real-time service 4.2 and 3.2 cm over 13:24 to 24:00, the only hours archived. Over the same hours the corrected clock is about 3.5 times noisier than CODE alone for GPS. The correlation of c0 with (GFZ minus CODE) is GPS −0.08, Galileo +0.01, and with (ESA minus CODE) GPS −0.18, Galileo +0.01. *Checked.*

**Why it works.** GFZ is the reference because the network's a priori clock is CODE's: against CODE the network would be measured partly against itself. A correction that carried clock information would correlate with CODE's error; c0 does not, and relative to its a priori it behaves as added noise of 6.3 cm (GPS).

**Pitfall.** Comparing rows with different hours (01:00 to 06:00 against 00:00 to 06:00, or against the real-time service's 13:24 to 24:00), or reading a standalone clock comparison as the user's precision: the user's phase term is phase bias minus clock, which the hold keeps at CODE's clock (Question 6.3).

**Takeaway.** The network does not yet improve on CODE's clock; its value today lies in the phase biases, the declared reference and the measured health.

*Source.* `results/t3/table_clock_report.tex`; `results/t3/results.md` lines 9 to 14, 100 to 113; `results/t3/clock_stats.csv`, rows `COD,GFZ,01-06`.

*Check.*

```
T3=D:/GnssProcessorSupport/presentation/status_update_2026-10-29/results/t3
grep "COD,GFZ,01-06" $T3/clock_stats.csv
```

**Answer 9.4** (question, p. 22)

**Strategy.** For each topic, find the question whose answer gives the evidence, state what that evidence shows, then state the nearest claim it does not support.

**Solution.**

(a) *Model and check.* May: user and network use one undifferenced, uncombined observation model, and the user one filter in every step; the rank gate found exactly the declared 2 directions (one clock per constellation) on 720 of 720 epochs, none withheld, on both reference cases (Question 3.5). May not: that the network uses the user's filter (it has its own information-form solve); that the check proves the choices harmless (it counts null directions, Question 4.3); that it covers each station's level (Question 4.5).

(b) *Reference choices.* May: they are declared in the correction file and every change of pivot is marked (Questions 5.2 and 5.3). May not: "a few" (226 rows and 120 prior entries, Question 4.3); all declared on real data (the Galileo pin pairs are not); that they keep a user's integers (Question 7.3).

(c) *Satellite clock.* May: its precision comes from CODE's final clock; the correction adds code noise and no clock information (Question 9.3); on simulated clock error the self-contained postures reach about 2 cm, unarmed (Question 6.4). May not: that it beats real-time clocks; that the network is free of outside clock products; that the quiet simulation tests the clock (Question 8.1). Without CODE's clock a user was 1.7 cm worse at the median, against an older GPS-only configuration (Question 6.5).

(d) *Integers.* May: on a simplified simulation a log-only search fixed 42 % of 13,138 double differences, none wrongly (Question 7.5). May not: integers in the corrections; integers on real data; producer-side fixing, whose standing verdict is DO NOT BUILD (`docs/network_ambiguity_resolution_design.md` lines 3 to 5).

(e) *Results.* May: on 8 May 2024, WTZZ00DEU 1.0 to 1.8 cm and the median of 16 further stations 3.6 to 5.4 cm in hours 2 to 6, without integers; over five days the median station 3.8 cm (mean 4.1 cm); on the quiet simulation within 0.9 cm of perfect corrections (Questions 8.4, 9.1 and 9.2). May not: the best station on every day; WTZZ00DEU as a user unrelated to the network; claims from the first hour.

**Why it works.** A claim is bounded by the conditions of its evidence: the data set, the hours, the configuration and the instrument. Each "may not" is the nearest claim that drops one of those conditions.

**Pitfall.** Rounding a scoped result into a general one: one day into every day, one simulation into the method, a float-solution search into published integers.

**Takeaway.** These boundaries were applied to the status report's wording on 24 October 2026.

*Source.* the answers named.

## How the answers were checked

On 24 October 2026, against branch `card/scb-clock-bed` and the runs named, each check below reproduced the answers it lists. Three statements gathered for an earlier version of the key failed their check and were removed.

| Answers | Check | Result |
|---|---|---|
| 2.1 to 2.6, 3.1, 3.2 | a numpy script that builds the worked example's systems | every number of those answers; ranks 5, 13 (18 and 26 unknowns) and 7 |
| 3.4, 3.5 | the parameter table of `runs/_rf0hv15_sim6`; `rf_null_space_read.py` | 280 shared columns; eigenvalues 4.036e-11 and −1.803e-10, weights 0.2040 and 0.1767; cutoff 1.0459e-6 |
| 4.2, 6.3 | `make_golden_config.py --verify`; the loader lines | "golden verify: PASS"; the refusal and warning as quoted |
| 4.4 | B records of G05 against the navigation file | $-3.2108400641$ m $= cT_{GD}$; ratio 1.646944 |
| 5.3 | the datum-step script | 31 satellites, 1.6278 m, 0.0027 m after the mean |
| 5.4 | `simnet_amb_integer.py --ref-free` | 0.304 and 0.122 cycle |
| 6.4, 8.3 | `simnet_clock_truth_grade.py` on the card SCB cases and the baseline | every number of those answers |
| 7.5 | `simnet_wire_integer_consistency.py`; `simnet_producer_ar_grade.py` | 0.252 to 0.260 against 0.250; 5572 of 5572 |
| 9.1 to 9.3 | `rapid_arm.py score`; `gcensus.py`; the five-day script; `clock_stats.csv` | the hourly medians; −0.0198 m; 0.038, 0.0406, 0.018 m; 1.20, 2.23 cm |
| all derive answers | one script with their arithmetic | as given |
| removed | "the 16 further stations have no co-located network station" | KOKB00USA and network station KOKV00USA share a RINEX header position |
| removed | "integer fixing equals the float within 0.1 mm at ALGO00CAN" | the cited line gives another comparison |
| removed | "code-bias records agree to 3e-11 m" | from a scratch script; replaced by the run's own count |

## References


[1] P. J. de Jonge. "A processing strategy for the application of the GPS in networks". PhD thesis. Delft, The Netherlands: Delft University of Technology, 1998.

[2] D. Odijk et al. "On the estimability of parameters in undifferenced, uncombined GNSS network and PPP-RTK user models by means of S-system theory". In: *Journal of Geodesy* 90.1 (2016), pp. 15–44.

[3] P. J. G. Teunissen. "Generalized inverses, adjustment, the datum problem and S-transformations". In: *Optimization and Design of Geodetic Networks* (1985), pp. 11–55.